Chapter 10
Sound, Speech, and Hearing

Abstract The basic physics of sound is first developed, including that of sound wave propagation in air and at interfaces, and of sound in resonant cavities, and these concepts are then applied to speaking and hearing. The structure and function of the relevant body components for speaking and hearing are presented, followed by a discussion of how the body produces sound and senses it. Other types of body sounds are then overviewed.

In this chapter we will ask: *How do we speak? How do we hear?* To answer these questions we will have to probe into the physics of sound waves, which is called acoustics, because speaking is creating sound and hearing is detecting sound. We will first review the basic physics of sound waves and will then examine the physics of speaking and hearing by humans in more detail.

Sound is also important in medical diagnostics. Physicians use stethoscopes to listen to sounds in the body (*auscultation*; see the end of this chapter) and they tap on parts of the body, as one would a drum, as a percussive diagnostic. Another important medical diagnostic using sound is *ultrasonic mapping* or *ultrasonography*. It is a nondestructive imaging method that makes use of sound at frequencies ($\sim$1–10 MHz), way above our hearing range (20–20 kHz), that provides images with the very useful spatial resolution of $\sim$1 mm [51, 72]. Waves are sent to an object and reflected, with the delay time between the transmission of the probe beam and the arrival of the reflected acoustic pulses at the detector giving the relative location of the object. For example, in analyzing the heart the use of a scanned single beam gives valuable, yet limited information, such as the wall thickness and chamber diameters (*M-mode echocardiography*), while the use of multiple beams transmitted through a wide arc provides two-dimensional images of the heart (*2-D echocardiography*). The shifting of the acoustic frequency when the ultrasound reflects from a moving target (*the Doppler effect*) is the basis for measuring blood flow direction, turbulence, and speed (*Doppler ultrasonography*). (See Problems 10.22 and 10.23, and Fig. 10.56 below.)

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10.1 The Physics of Sound Waves

Sound is a compressional wave in a gas, liquid, or solid. A wave is a periodic disturbance that travels in space, say in the z direction. It is periodic in space, which means that at any given time t, the disturbance repeats periodically with z, as in Fig. 10.1. It is periodic in time, which means that at any given position z, the disturbance repeats periodically with time t. The disturbance travels with a speed v, the speed of sound, so from time t_1 to time t_2 the disturbance travels a distance $\delta z = z_2 - z_1 = v(t_2 - t_1)$. The quantity $z - vt$ does not change for the disturbance as it "travels" with the wave.

So far we have described the propagation of a disturbance by a generic type of wave, but have not specified what is being disturbed. In sound waves, these disturbances are local changes in pressure, $\delta P = P(z,t) - P_{\text{ambient}}$, mass density, $\delta\rho$ (or molecular density, δn), and displacement, δz, from their ambient values. Sound waves in gases and liquids are *longitudinal* or *compressional* in that these changes occur in the same direction as the wave propagates, here in the z direction. If you pluck a string, the wave propagates along the string, but the actual disturbance of

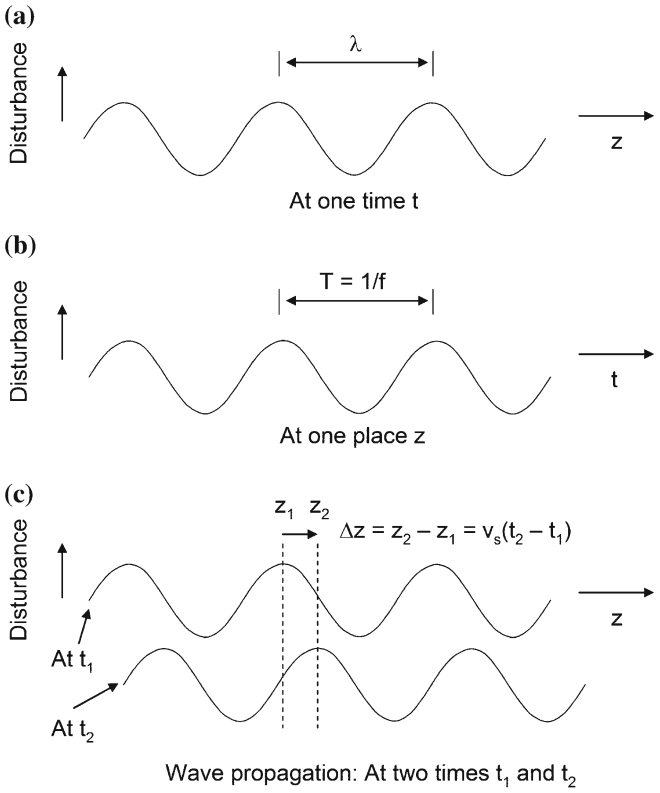


Fig. 10.1 Waves at **a** one time, **b** one place, and **c** two different times, showing wave propagation

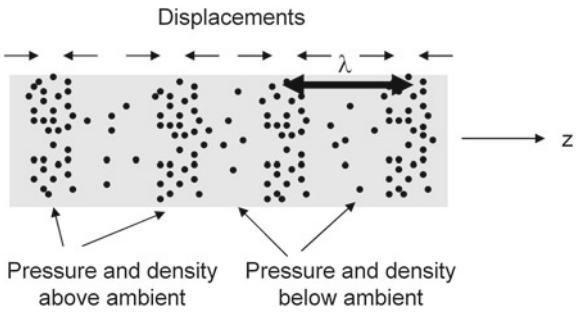


Fig. 10.2 Sound waves are compressional waves

the string is perpendicular to it, making it a *transverse* wave. (In solids, sound waves can be longitudinal or transverse.) Figure 10.2 shows the longitudinal motion of the molecules during a sound wave. These pressure and density variations are in phase with each other, meaning that they both increase (compression) or decrease (rarefaction) from the ambient values together. (The equations of state of materials, such as $P = nRT$ for ideal gases, usually show that density increases with pressure.) In contrast, the displacement of molecules is out of phase with density and pressure; the displacement spatially varies as a cosine wave if the density and pressure vary as sine waves. Where the density and pressure are at a maximum, the displacement is zero, but the displacement is positive just to the left and negative just to the right—which maximizes the density and pressure. Where the density and pressure are at a minimum, the displacement is also zero, but the displacement is negative just to the left and positive just to the right. This is seen in Fig. 10.2.

We have chosen so far to examine the simplest waves, for which the periodic disturbance varies as a sine function (or equivalently a cosine function). While sound waves from speaking are complex sums of such waves at different frequencies, to a very good approximation we can examine the physics of each frequency by itself and sum the effects. (This is called linearity or linear superposition.)

10.1.1 The Speed and Properties of Sound Waves

Sound waves move at a speed v_s that is determined by the properties of the medium. In general the sound speed is

$$v_s = \sqrt{\frac{C}{\rho}}, \tag{10.1}$$

where C is a constant describing the stiffness of the material (when there is no heat flow, which are "adiabatic" conditions) and ρ is the mass density. In solids, this stiffness constant can depend on the direction the sound wave propagates. It equals Young's modulus Y for the propagation of compressional waves down a rod that is

much longer than it is wide. For steel, $v_s = 5{,}960\,\text{m/s}$. In fluids (liquids and gases), C is the (adiabatic) bulk modulus B, which describes how much pressure is needed to achieve a given fractional decrease in volume. In gases, $B = \gamma P$, where γ is the ratio of the specific heats at constant pressure (c_p) and volume (c_v). (The ratio c_p/c_v ranges from 1, for very large molecules, to 5/3, for an ideal monatomic gas, and is 1.4 for air, which is composed mostly of diatomic gases.) Consequently, the speed of sound in gases is

$$v_s = \sqrt{\frac{\gamma P}{\rho}} = \sqrt{\frac{\gamma RT}{m}}, \tag{10.2}$$

where R is the constant in the ideal gas law (7.2) ($R = 8.31\,\text{J/mol-K}$), and m is the molecular mass. The speed of sound in air is 343 m/s (at 20 °C), which is 15× slower than that in steel, while in water it is 1,482 m/s (see Problems 10.4 and 10.5).

Waves are periodic in space (at one t) and repeat with a spatial periodicity called the wavelength λ (Fig. 10.1). They are also periodic in time (at one z) with a temporal periodicity called the period T, which corresponds to a frequency f with $f = 1/T$ (Fig. 10.1). Mathematically, a disturbance moving to the right (towards larger z) can be expressed as $g(z - v_s t)$—for any wave of functional form g, because $z - v_s t$ remains constant as the wave moves to the right at speed v_s. One example of this is $\cos(z - v_s t)$. Similarly, a disturbance moving to the left (towards smaller z) can be expressed as $g(z + v_s t)$, because $z + v_s t$ remains constant as the wave moves to the left.

Frequency has units of cycles per second (cps) = Hertz (Hz). Sometimes we will use the radial frequency ω, which has units of rad/s or just 1/s, and which is related to the frequency by $\omega = 2\pi f$. The wavelength, frequency, and speed of a wave are interrelated by

$$v_s = \lambda f. \tag{10.3}$$

So, low-frequency waves have long wavelengths, while high-frequency waves have short wavelengths. Using the speed of sound in air, $f = 1{,}000\,\text{Hz}$ sound waves in air have a wavelength of $0.34\,\text{m} \simeq 1\,\text{ft}$. For some types of waves, v_s depends on f, and so $v_s = v_s(f)$, but it does not for sound in air. When sound waves travel from one medium to another, the frequency stays the same, but the wavelength changes with the change in sound speed (10.3). Note that for acoustic waves it is the disturbance that is propagating; the actual molecules do not physically travel with the disturbance and move very little.

10.1.2 Intensity of Sound Waves

The intensity I of a sound wave is the energy carried by the wave per unit area and per unit time (in units of J/m^2-s or W/m^2). At a distance R from an isotropic source of average acoustic power P_{power}, the intensity is

$$I = \frac{P_{\text{power}}}{4\pi R^2}. \tag{10.4}$$

The acoustic intensity is also equal to the kinetic energy of the wave per unit volume, $\rho u_{\text{max}}^2/2$, times the wave speed, v_{s}, or

$$I = \frac{1}{2}\rho u_{\text{max}}^2 v_{\text{s}}, \tag{10.5}$$

where $u_{\text{max}} = (\delta z_{\text{max}})\omega$ is the maximum speed of the molecules for a maximum displacement δz_{max} during the disturbance. Therefore, we see that

$$I = \frac{1}{2}\rho v_{\text{s}}[(\delta z_{\text{max}})\omega]^2. \tag{10.6}$$

The *acoustic impedance* of a medium Z is given by the product of the mass density and sound speed for that medium, so

$$Z = \rho v_{\text{s}}. \tag{10.7}$$

Table 10.1 lists the mass density, sound speed, and acoustic impedance for air, water, fat, muscle, and several other body materials. The acoustic impedance for water, fat, and muscle are ~3,500× that for air. We will see later that this mismatch is responsible for the reflection of sound between air and these other media.

Table 10.1 Mass density, sound speed, and acoustic impedance

Material	ρ (kg/m^3)	v_{s} (m/s)	Z ($= \rho v_{\text{s}}$) (kg/m^2-s)
Air (20 °C)	1.20	343	413
Water	1.00×10^3	1,480	1.48×10^6
Fat	0.92×10^3	1,450	1.33×10^6
Muscle	1.04×10^3	1,580	1.64×10^6
Bone	2.23×10^3	3,500	7.80×10^6
Blood	1.03×10^3	1,570	1.61×10^6
Soft tissue (avg.)[a]	1.06×10^3	1,540	1.63×10^6
Lung	286	630	1.80×10^5

Using data from [24, 51]

[a]The soft tissue value is representative of the skin, kidney, liver, and the brain

For simplicity, we will relabel the gauge pressure δP ($= P - P_{\text{atmosphere}}$) as P for the rest of this chapter and call it the sound pressure. Moreover, P will actually denote the maximum pressure in the cycle, $P_{\max}$. The magnitude of the maximum pressure variation in the sound wave is related to this maximum (out of phase) displacement by

$$|P| = (\rho v_s)\omega|\delta z_{\max}| = Z\omega|\delta z_{\max}|. \tag{10.8}$$

We can understand this because pressure is the force per unit area and force is the change of linear momentum with time, and so pressure is the change in linear momentum per unit area per unit time. Using (10.8), the acoustic intensity is

$$I = \frac{1}{2}Z(\delta z_{\max})^2\omega^2 = \frac{P^2}{2Z}. \tag{10.9}$$

We can present the sound intensity I in units of W/m^2 or any other equivalent units. It is also very common to characterize the sound intensity in a more physiologically-based manner, in which I is referenced to $I_{\text{ref}} = 10^{-12}$ W/m^2; I_{ref} is a sound intensity that is barely audible at 3,000 Hz. Because sound intensities in our everyday experience can be many orders of magnitude larger than this reference intensity, we usually use a logarithmic scale—referenced to base 10—to characterize I. In units of bels, named after Alexander Graham Bell, I (in bels) $= \log_{10}(I/I_{\text{ref}})$. It is, in fact, standard to use a finer scale in tenths of bels, called decibels or dB, with

$$I\,(\text{in dB}) = 10\log_{10}\frac{I}{I_{\text{ref}}}. \tag{10.10}$$

For example, for $I = 10^{-8}$ W/m^2 $= 10^4 I_{\text{ref}}$ we see that $I\,(\text{in dB}) = 10\log_{10}(10^4) = 10 \times 4 = 40$, or 40 dB.

The dB scale is also used to denote the relative magnitude of intensities, such as that of I_2 relative to I_1. With

$$I_1(\text{in dB}) = 10\log_{10}\frac{I_1}{I_{\text{ref}}} \quad \text{and} \quad I_2(\text{in dB}) = 10\log_{10}\frac{I_2}{I_{\text{ref}}} \tag{10.11}$$

we see that

$$I_2(\text{in dB}) - I_1(\text{in dB}) = 10\log_{10}\frac{I_2}{I_{\text{ref}}} - 10\log_{10}\frac{I_1}{I_{\text{ref}}} = 10\log_{10}\frac{I_2}{I_1}. \tag{10.12}$$

So, a 20 dB increase in sound intensity indicates a factor of $10^2 = 100$ increase in I. Because the dB scale is also used to indicate relative magnitudes, this absolute acoustic unit in (10.10) is often denoted as dB SPL (sound pressure level). Table 10.2 shows the acoustic intensities of common sounds. Normal background noise is 50–60 dB SPL. Normal conversation is 60–70 dB SPL. Speech is around 70–80 dB SPL at 1 m. The threshold of pain occurs at about 120 dB SPL. Windows break at about

Table 10.2 Typical sound intensities

	Intensity (W/m^2)	Intensity level (dB SPL)
Sound barely perceptible, human with good ears	10^{-12}	0
Human breathing at 3 m	10^{-11}	10
Whisper at 1 m, rustling of leaves, ticking watch	10^{-10}	20
Quiet residential community at night, refrigerator hum	10^{-8}	40
Quiet restaurant, rainfall	10^{-7}	50
Normal conversation at 1 m, office, restaurant	10^{-6}	60
Busy traffic	10^{-5}	70
Loud music, heavy traffic, vacuum cleaner at 1 m	10^{-4}	80
Loud factory	10^{-3}	90
Fast train, pneumatic hammer at 2 m, disco, blow dryer	10^{-2}	100
Accelerating motorcycle at 5 m, chainsaw at 1 m	10^{-1}	110
Rock concert, jet aircraft taking off at 100 m	$1 = 10^{0}$	120
Jackhammer	10^{1}	130
Shotgun blast, firecracker	10^{2}	140
Jet engine at 30 m	10^{3}	150
Rocket engine at 30 m	10^{6}	180

163 dB SPL. Shock waves and sonic booms cause levels of about 200 dB SPL at a distance of 330 m.

Apollo astronauts were exposed to very loud sounds during liftoff, over 85 dB SPL for about 80 s after liftoff—with maximum levels below 100 Hz [25, 37]. About 60 s after launch the sound in the crew area was about 123 dB over a wide frequency range, but astronauts heard much less intense sounds (because they wore helmets and suits), from 119 dB at 63 Hz to 89 dB at 2,000 Hz. On the gantry, 10 m above ground, the maximum levels were 158–168 dB SPL from 2 to 2,000 Hz and then down to 152 dB at 8,000 Hz [37, 39]. The levels were 12–31 dB lower 400 m from the gantry.

Equation (10.9) relates the sound intensity and sound pressure by $I = P^2/2Z$. For the reference intensity $I_{\text{ref}} = P_{\text{ref}}^2/2Z$, and with $I_{\text{ref}} = 10^{-12}\,\text{W/m}^2$ and $Z = 413\,\text{kg/m}^2\text{-s}$ for air, we see that the reference pressure $P_{\text{ref}} = 2.9 \times 10^{-5}\,\text{N/m}^2$ (Pa). Dividing these two relations for intensity and pressure gives

$$\frac{I}{I_{\text{ref}}} = \frac{P^2}{P_{\text{ref}}^2} \tag{10.13}$$

and using (10.10) gives

$$I\,(\text{in dB SPL}) = 20 \log_{10} \frac{P}{P_{\text{ref}}}. \tag{10.14}$$

Sometimes a distinction is made between the expression for the intensity, (10.10), which gives the intensity level (IL) in units of dB IL, and that from (10.14) for pressure, which gives the sound pressure level in dB SPL, but we will not make such a distinction here and the units will be called dB SPL. Generalizing (10.12) gives

$$I_2(\text{in dB SPL}) - I_1(\text{in dB SPL}) = 10\log_{10}\frac{I_2}{I_1} = 20\log_{10}\frac{P_2}{P_1}. \tag{10.15}$$

When the sound intensity increases by 60 dB, I increases by 10^6 and P increases by 10^3.

There is an analog to (10.9) in electronics with power $P_{\text{elect}} = V_{\text{elect}}^2/2R_{\text{elect}}$, where V_{elect} is the voltage drop and R_{elect} is the resistance. (The factor of 2 in the denominator is present for sinusoidal voltages and absent for DC voltages.) As in acoustics, the dB scale is commonly used in electronics analogous to (10.10), (10.12), and (10.15).

Absorption of Sound

Within a given medium the sound wave can be attenuated by absorption and scattering. In scattering, part of the propagating beam is redirected into many directions, without being absorbed. The amplitude of the acoustic wave (A = magnitude of the pressure $\delta P_{\max}$ or displacement $\delta z_{\max}$) decreases exponentially with the distance the wave propagates, z. This can be written as

$$A(z) = A(z=0)\exp(-\gamma_{\text{sound}} F z), \tag{10.16}$$

where γ_{sound} is the absorption coefficient and F is the frequency dependence. For pure liquids, $F = f^2$ in Hz^2 and for soft tissues, $F \sim f$ in Hz. Typical values of γ_{sound} are given in Table 10.3 for body tissues and in Table 10.4 for other materials. Using (10.13), the acoustic power or intensity varies as

$$I(z) = I(z=0)\exp(-2\gamma_{\text{sound}} F z). \tag{10.17}$$

This variation also applies to the absorption of light in media and is called Beer's Law. In Beer's Law jargon the intensity absorption coefficient $2\gamma_{\text{sound}} F$ in (10.17) is called α_{sound}, and so, for instance, $\alpha_{\text{sound}} = 2\gamma_{\text{sound}} f$ for tissue. Beer's Law for light is usually expressed as:

$$I(z) = I(z=0)\exp(-\alpha_{\text{light}} z). \tag{10.18}$$

In analyzing the measured fraction of the intensity of sound or light transmitted through a medium, all of the loss mechanisms must be considered: the reflection from interfaces (see below), absorption, and scattering. Because the wavelengths of sound waves are of the same order of magnitude as everyday objects, such as windows, pipes, heads, and so on, the diffraction and interference of sound waves is very common.

Table 10.3 Amplitude absorption coefficient γ_{sound} for tissues

Tissue	γ_{sound} (s/m)
Aqueous humor	1.1×10^{-6}
Vitreous humor	1.2×10^{-6}
Blood	2.1×10^{-6}
Brain (infant)	3.4×10^{-6}
Abdomen	5.9×10^{-6}
Fat	7.0×10^{-6}
Soft tissue (average)	8.3×10^{-6}
Liver	1.0×10^{-5}
Nerves	1.0×10^{-5}
Brain (adult)	1.1×10^{-5}
Kidney	1.2×10^{-5}
Muscle	2.3×10^{-5}
Crystalline eye lens	2.6×10^{-5}
Bone	1.6×10^{-4}
Lung	4.7×10^{-4}

Using data from [24]
It is multiplied by the frequency f (in Hz) to obtain the amplitude absorption coefficient per unit length

Table 10.4 Amplitude absorption coefficient γ_{sound} for fluids

Fluids	γ_{sound} (s^2/m)
Water	2.5×10^{-14}
Castor oil	1.2×10^{-11}
Air (STP)	1.4×10^{-10}

Using data from [24]
It is multiplied by f^2, where f is the frequency (in Hz), to obtain the amplitude absorption coefficient per unit length. STP is standard temperature and pressure

Components of Impedance (Advanced Topic)

The concept of impedance is more generally used to assess the characteristics of a medium that "opposes" energy flow in a system. It is also used to characterize electrical components and circuits, where the electrical resistance is the real part of the impedance and the capacitance and inductance of a system correspond to the imaginary part of the electrical impedance. Analogous concepts are used in optics to characterize the transmission of light in media. In acoustics (and other areas), the reciprocal of the *impedance* Z, is called the *admittance* $Y = 1/Z$, which describes the ease of energy flow. *Immittance* refers to either the impedance or admittance, and is used as a general term to describe how well energy flows in a medium.

The impedance is generally a complex parameter. The real part of Z is the *resistance* R. Out of phase to the resistance by 90° (or $\pi/2$ rad) are the *mass (positive) reactance* (X_{m}), which is proportional to frequency f, and the *stiffness (or negative) reactance* X_{s}, which is inversely proportional to f; X_{m} and X_{s} are 180°

(or π rad) out of phase to each other. Overall, $Z = R + i(X_m - X_s)$. The net reactance is $X_{net} = | X_m - X_s |$. The magnitude of the impedance is related to these component parts by

$$| Z | = \sqrt{R^2 + X_{net}^2} \, . \tag{10.19}$$

Similarly, the admittance $Y = G + i(B_m - B_s)$. The real part G is the *conductance*, and the imaginary parts B_m and B_s are the *mass susceptance* and *stiffness (compliant) susceptance*, respectively. (When the impedance has only a resistive term, the conductance is $G = 1/R$.) The net susceptance is $B_{net} = | B_m - B_s |$. The components of the admittance are related to the impedance terms by $G = R/(R^2 + X_{net}^2)$, $B_m = -X_m/(R^2 + X_{net}^2)$, and $B_s = -X_s/(R^2 + X_{net}^2)$. The magnitude of the admittance is related to these component parts by

$$| Y | = \sqrt{G^2 + B_{net}^2} \, . \tag{10.20}$$

Problems 10.26–10.29 address these and other relations involving impedances.

This discussion can apply to the impedance in acoustics, electronics, or flow (Appendix D). For example, this terminology can be applied to the analog in mechanics of a mass attached to a spring that slides on a rough surface. The friction due to sliding on the rough surface is the resistance. The inertia due to the mass is the mass reactance and the resistance to movement due to the spring is the stiffness reactance. When used in audiology and other aspects of sound, these parameters are also characterized by the term "acoustic," such as acoustic impedance, acoustic admittance, acoustic resistance, mass (positive) acoustic reactance, and so on. To avoid confusion we could use subscripts, such as R_{acoust} for acoustic resistance, and so on.

The units of impedance Z is ohms (Ω), just as in electronics. The units of admittance Y is the 1/ohm = 1 mho. R, X_m, and X_s all have units of ohms, and G, B_m, and B_s all have units of mhos. In audiology the admittance values are small and the more useful unit is the millimho or mmho.

The Impedance in a Harmonically Driven System (Advanced Topic)

We now examine the concept of impedance and derive (10.19) by considering the motion of a body of mass m attached to a spring of force constant k. It is subject to a viscous, frictional force, characterized by η, and a driving term, $F_A \exp(i\omega t)$. This motion is described by

$$m\frac{d^2x}{dt^2} + \eta\frac{dx}{dt} + kx = F_A \exp(i\omega t). \tag{10.21}$$

The driving term has a real part $F_A \cos(\omega t)$, because $\exp(i\omega t) = \cos(\omega t) + i \sin(\omega t)$. This model is similar to the Voigt model in Chap. 4 with $F = \eta \, dx/dt + kx$ (from (4.57)). (It can be represented by the model presented later in Fig. 10.13, with the force driving term acting laterally on the mass.)

Substituting a potential solution $x = x_0 \exp(i\omega t)$ in this equation, gives the ("particular" or steady state) solution

$$x = \frac{F_\mathrm{A}}{-m\omega^2 + \mathrm{i}\eta\omega + k} \exp(i\omega t). \tag{10.22}$$

(See Appendix C.) Differentiating this, the speed of the body is

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{\mathrm{i}\omega F_\mathrm{A}}{-m\omega^2 + \mathrm{i}\eta\omega + k} \exp(\mathrm{i}\omega t) \tag{10.23}$$

$$= \frac{F_\mathrm{A}}{\eta + \mathrm{i}m\omega + k/\mathrm{i}\omega} \exp(\mathrm{i}\omega t). \tag{10.24}$$

The ratio of the driving force, $F_\mathrm{A} \exp(i\omega t)$, to the speed, $\mathrm{d}x/\mathrm{d}t$, is the mechanical impedance Z

$$\frac{F_\mathrm{A} \exp(\mathrm{i}\omega t)}{\mathrm{d}x/\mathrm{d}t} = \eta + \mathrm{i}m\omega + k/\mathrm{i}\omega = Z. \tag{10.25}$$

This impedance is sometimes expressed as

$$Z = R + \mathrm{i}\omega M + S/\mathrm{i}\omega = R + \mathrm{i}(\omega M - S/\omega). \tag{10.26}$$

Here, $R = \eta$ provides the resistance, $M = m$ is the inertial term (or inertance), and $S = k$ is the stiffness. In terms of the *mass (positive) reactance* (X_m), which is proportional to frequency f, and the *stiffness (or negative) reactance* X_s, which is inversely proportional to f, we can say

$$Z = R + \mathrm{i}X_\mathrm{m} + X_\mathrm{s}/\mathrm{i} = R + \mathrm{i}(X_\mathrm{m} - X_\mathrm{s}), \tag{10.27}$$

with $X_\mathrm{m} = \omega M = 2\pi f M$ and $X_\mathrm{s} = S/\omega = S/2\pi f$. Equation (10.19) follows from the magnitude of a complex number.

10.1.3 What Happens When Sound Travels from One Medium to Another?

Sound transmission from one medium to another is very important in hearing, because sound is transmitted from the air in the auricle and ear canal into the middle ear and then into the inner ear. This can be understood by seeing what happens to a sound wave incident on the planar interface between two different semi-infinite media, such as media 1 and 2 in Fig. 10.3.

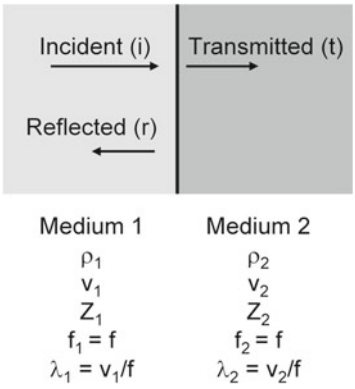


Fig. 10.3 Schematic of acoustic wave transmission and reflection

This sound wave travels in medium 1 with intensity I_i and pressure P_i incident on the interface. The part that is reflected back into the same medium has intensity I_r and pressure P_r, and the part that is transmitted into medium 2 has intensity I_t and pressure P_t. These pressures are related by "matching the boundary conditions" at the interface, as we will see very soon. As stated earlier, the frequency is the same in both media, but the wavelength changes with the change in sound speed by (10.3), so $\lambda_1 = v_1/f$ and $\lambda_2 = v_2/f$. These frequencies need to be the same because the sinusoidal oscillations of pressure and matter movement on both sides of the interface must always match. The magnitudes of the pressures must match so there is no net force on the interface. This gives

$$P_\mathrm{i} + P_\mathrm{r} = P_\mathrm{t}. \tag{10.28}$$

The motion at the interface caused by all three waves must match. The speed of this lateral displacement is $|\omega(\delta z)|$ and for a wave initially moving to the right it is positive for the incident and transmitted waves and negative for the reflected wave. Using (10.8), $\delta z \sim P/(Z\omega)$, we know that $|\omega(\delta z)| \sim P/Z$, and so matching displacement gives

$$\frac{P_\mathrm{i}}{Z_1} - \frac{P_\mathrm{r}}{Z_1} = \frac{P_\mathrm{t}}{Z_2}. \tag{10.29}$$

Solving these two equations gives

$$\frac{P_\mathrm{r}}{P_\mathrm{i}} = \frac{Z_2 - Z_1}{Z_1 + Z_2} \tag{10.30}$$

$$\frac{P_\mathrm{t}}{P_\mathrm{i}} = \frac{2Z_2}{Z_1 + Z_2}. \tag{10.31}$$

Using (10.9) and (10.30), the fraction of intensity that is reflected is

$$R_{\text{refl}} = \frac{I_r}{I_i} = \frac{P_r^2/2Z_1}{P_i^2/2Z_1} = \frac{P_r^2}{P_i^2} \tag{10.32}$$

$$= \left(\frac{Z_2 - Z_1}{Z_1 + Z_2}\right)^2 = \left(\frac{1 - Z_2/Z_1}{1 + Z_2/Z_1}\right)^2. \tag{10.33}$$

Using (10.31), the fraction of intensity that is transmitted is

$$T_{\text{trans}} = \frac{I_t}{I_i} = \frac{P_t^2/2Z_2}{P_i^2/2Z_1} = \frac{Z_1}{Z_2}\frac{P_t^2}{P_i^2} \tag{10.34}$$

$$= \frac{Z_1}{Z_2}\left(\frac{2Z_2}{Z_1 + Z_2}\right)^2 = \frac{4Z_2/Z_1}{(1 + Z_2/Z_1)^2}. \tag{10.35}$$

We see that $I_i = I_r + I_t$ and $R_{\text{refl}} + T_{\text{trans}} = 1$, which means that sound energy (and intensity) is conserved. Also, the reflected fraction is large (and approximately equal to 1) when Z_2/Z_1 is either much larger or smaller than 1, i.e., when there is a very large acoustic impedance mismatch. We will see this means the large mismatch of acoustic impedances between air and any solid or liquid medium (Table 10.1) has important consequences in the performance of the ear. The term "to match impedance" means to minimize R_{refl}, and this is accomplished by making Z_1 and Z_2 approximately equal (10.33). Examples of this mismatch in the body are given in Table 10.5. Most sound incident on the body is reflected. Little sound is reflected between soft tissues (brain, skin, kidney, liver, and so on) and between soft tissues and blood. There is significant reflection at interfaces of soft tissues with the lungs and with bones.

Table 10.5 Representative fractions of reflected and transmitted acoustic energy at tissue interfaces

Tissue interface	Reflected fraction (in %)	Transmitted fraction (in %)
Water/soft tissue	0.23	99.77
Fat/muscle	1.08	98.92
Bone/muscle	41.23	58.77
Soft tissue/bone	43.50	56.50
Bone/fat	48.91	51.09
Soft tissue/lung	63.64	36.36
Air/muscle	98.01	1.99
Air/water	99.89	0.11
Air/soft tissue	99.90	0.10

Using data from [24, 51, 74]

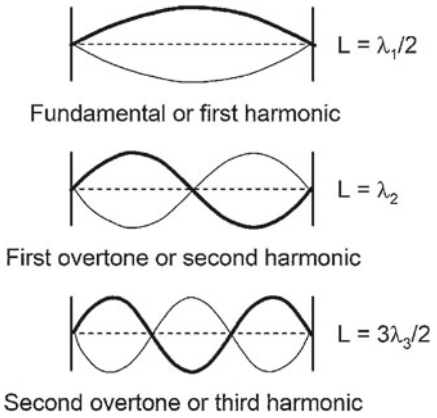


Fig. 10.4 Lowest three modes on a string of length L, each shown at times with maximum and zero excursions (Based on [28])

10.1.4 Resonant Cavities

Many properties of sound waves depend on the medium the wave propagates in and on the characteristics of the enclosure. This is important in the production of sound—as in human speaking, musical instruments, megaphones, and sound produced in echo chambers—and in the collection of sound waves—as in human hearing in the outer ear. One way to analyze these properties is to consider the properties of sound waves in resonant cavities; more generally, these properties of *resonant cavities* can be applied to any waves in any resonant structure.

What is a resonant structure? Let us consider a wave on a string of length L. The string can be plucked so a localized wave can propagate along the length of the string. Alternatively, one can pluck the string so a (transverse wave) oscillates everywhere periodically. If the string is rigidly fixed at both ends, the lowest order periodic motion (or mode—for which the transverse displacement at any point along the string varies as $\cos \omega t$) is one with a half wave between the two ends (Fig. 10.4). Here $\lambda/2 = L$, so the wavelength $\lambda = 2L$. The "boundary conditions" for this motion are zero transverse motion at each end; such places with no motion are called *nodes*. Higher order modes with higher integral numbers of half wavelengths are possible. In general, for n half wavelengths, the wavelength is $n(\lambda_n/2) = L$, and so the resonant wavelengths are

$$\lambda_n = \frac{2L}{n} \qquad \text{with } n = 1, 2, 3, \ldots . \tag{10.36}$$

Since $v_s = \lambda f$, the resonant or mode frequencies are

$$f_n = \frac{v_s}{\lambda_n} = n\frac{v_s}{2L}, \tag{10.37}$$

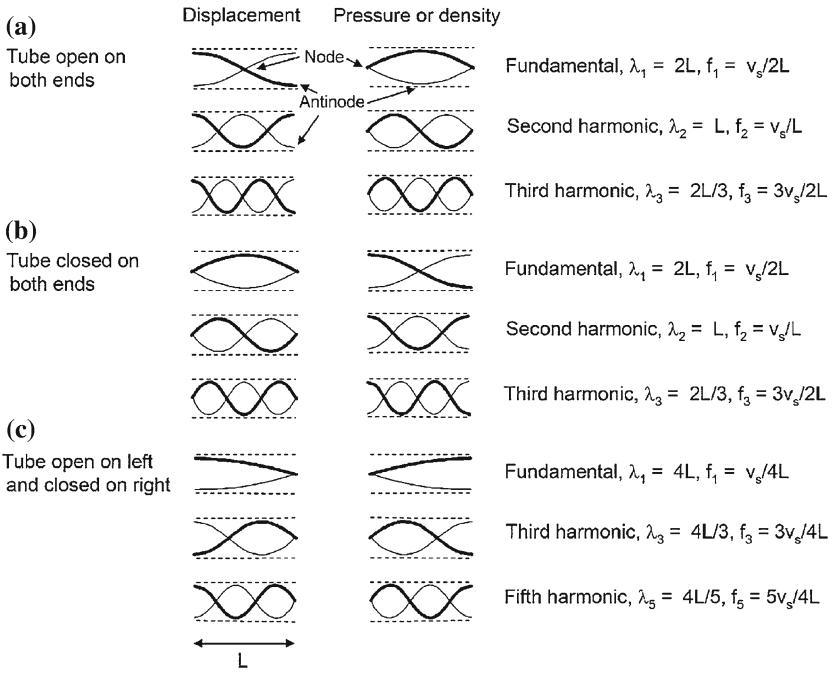


Fig. 10.5 Wave modes in a tube, for tubes **a** open on both ends, **b** closed on both ends, and **c** open on the *left side* and closed on the *right side*. The mode displacements of air are shown on the *left* for the first overtone or fundamental mode and for the next two overtones, and the corresponding changes in pressure and density for these modes are shown on the *right* (Based on [28])

which is $v_s/2L$, $2(v_s/2L)$, $3(v_s/2L)$, … for $n = 1, 2, 3, \ldots$; these are also called the *fundamental frequency*, the *first harmonic*, the *second harmonic*, etc. The mode frequencies in this case are equally spaced by $v_s/2L$.

If one of the two ends of the massless string is not fixed, but is free to move (only transverse to string axis), this different boundary condition leads to a different set of resonant wavelengths and frequencies. At the fixed end, the transverse displacement is still zero, which is a node. At the free end, the transverse displacement now has maximum magnitude (and zero slope)—and this is called *antinode* an, as in Fig. 10.5. The lowest order mode now has a quarter wavelength between the two ends separated by the string length, and so $\lambda/4 = L$ and the wavelength $\lambda = 4L$. The next lowest order mode has an extra half wavelength cycle with $3\lambda/4 = L$, so the wavelength $\lambda = 4L/3$, etc. Now the resonant condition is $(m/2 + 1/4)\lambda_m = L$, so the resonant wavelengths are

$$\lambda_m = \frac{2L}{m + 1/2} \qquad \text{with } m = 0, 1, 2, 3, \ldots. \tag{10.38}$$

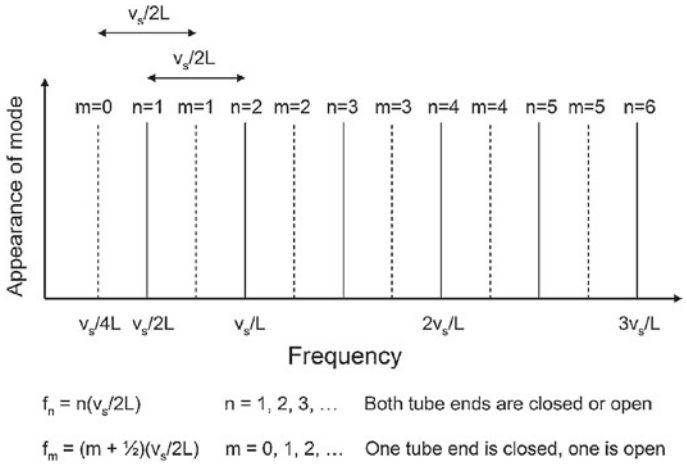


Fig. 10.6 Mode frequencies for a tube closed on both ends, open on both ends, or closed on one end and open on the other

The mode frequencies are

$$f_m = \frac{v_s}{\lambda_m} = \left(m + \frac{1}{2}\right)\frac{v_s}{2L}, \tag{10.39}$$

which is $v_s/4L$, $3v_s/4L$, $5v_s/4L$, … for $m = 0, 1, 2, \ldots$. Although the mode frequencies are different from the previous case, they are still equally spaced by $v_s/2L$. The lowest frequency is now $v_s/4L$ and only odd harmonics are present. Figure 10.6 shows the mode frequencies for both boundary conditions.

Hollow tubes are good first-order models for the tubes in the body where longitudinal sound waves propagate (similar to pipes in a pipe organ) (Fig. 10.6). If the tube of length L is open at both ends, the pressure is fixed at the ambient pressure at each end and $P = 0$ at the ends. This is formally equivalent to the string with fixed ends, so the resonant wavelengths and frequencies are given by (10.36) and (10.37). The density oscillation is also zero at each end (nodes) and the molecular displacement is of maximum magnitude at each end (antinodes). If the tube is open at one end and closed at the other, P is still zero at the open end (a node). At the closed end, the lateral displacement is fixed at zero (a node), while the pressure and density changes have maximum magnitude (an antinode). This is formally equivalent to the string fixed at one end and free at the order, so the resonant wavelengths and frequencies are given by (10.38) and (10.39). (The resonant wavelengths and frequencies for a tube closed at both ends are the same as for one that is open at both ends, but the identifications of nodes and antinodes are interchanged.) This is the reason why the tube length and boundary conditions are important in determining these resonances, and why these resonances can be changed by varying the tube conditions (length, varying cross-sectional area and shape)—as can happen in the region

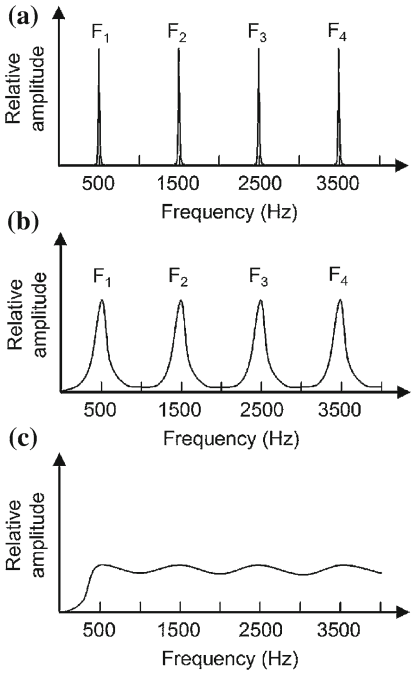


Fig. 10.7 **a** Loss-free (infinite Q), **b** moderate loss (moderate Q), and **c** very lossy (very low Q) transmission resonances for a tube (Based on [69])

between the vocal cords and the mouth opening. For details about acoustic resonances see [53, 54].

The mathematics of resonant excitation of an acoustic medium are the same as those used in Chap. 8 to analyze (8.48) and (8.50), and are due to acoustic losses. This damping leads to the finite widths of the resonances for acoustic energy, which are $\sim\gamma/2\pi$, in Hz, between the points with half the maximum response. Figure 10.7 compares the unrealistically narrow resonances to the more realistic, broader, damped resonances. The number of cycles that an excitation lasts during its decay (to $1/e$ of the initial energy) is $Q/2\pi$, where the quality factor $Q = \omega/\gamma = 2\pi f/\gamma$, where ω and f are the resonant frequencies. (See Appendix D for more details.)

10.2 Speech Production

10.2.1 Types of Sounds

Human speech is made by air from the lungs as it passes through the *trachea* (windpipe), *larynx* (which houses the *vocal folds or cords*), and *pharynx* (throat), through the mouth and nasal cavities, and then out of the mouth and nose (Fig. 9.1). For general sources about speech see [6, 7, 16, 17, 23, 29, 31, 43, 66, 69, 76].

The vocal folds (or vocal cords) are folds of ligament extending on either side of the larynx, with the space in between them called the *glottis*. The structure above the larynx is called the *vocal tract*. In adult females, the average length of the pharynx is 6.3 cm and that of the oral cavity is 7.8 cm, for a total vocal tract length of 14.1 cm. In adult males, the average length of the pharynx is 8.9 cm and that of the oral cavity is 8.1 cm, for a total vocal tract length of 16.9 cm.

Why can we vocalize distinct sounds well? One reason is that we are able to modify our very round tongues and the other parts of our vocal tract rapidly and precisely [43, 44]. Another is that our horizontal vocal tracts (above the tongue) and vertical tracts (below the tongue) are about equally long (Fig. 9.1), unlike those tracts in primates. Our horizontal tracts are shorter than in primates because our faces and jaws protrude less and our vertical tubes are longer because our larynxes are lower in the neck because they are suspended from the hyoid, a small floating bone at the base of the tongue. On the downside, with the larynx so low, there is an open region where food and air travel before entering the esophagus or trachea, so with this design we are susceptible to choking on food. (In other mammals, the epiglottis and soft palate touch and there are two tubes, an inner one for air flow and an outer one for food and fluid transport.)

The shape of the vocal tract can be varied by moving the soft palate, tongue, lips, and jaw, and these are the *articulators*. Adjusting the vocal tract to produce speech sounds is called *articulation*. The basic elements of speech are classified as the (1) phonemes (the basic sounds), (2) phonetic features (how the sounds are made), and (3) the acoustic signal (the acoustic nature of the sounds). We will first discuss the phonemes and the acoustic signal, and then describe how we make sound.

Phonemes are the shortest segments of speech, which in General American English are the 14 vowel sounds and 24 consonant sounds. Each phoneme is produced by distinctive movements of the vocal tract, which are the *phonetic features of speech*. Vowels are produced with the vocal tract relatively open, with different shapes of the opening. Consonants are produced by a constriction or closing of the vocal tract. The production of phonemes is characterized by three phonetic features (a) voicing, (b) place of articulation, and (c) manner of articulation.

Sounds produced as air rushes though the vibrating vocal folds in the larynx (Fig. 9.1) are called *voiced sounds*. All vowels are voiced sounds. (Touch the middle of your throat as you speak them.) Many consonants are also voiced, such as the "d," "m," "w," and "v" sounds. (We will not use the standard formal notation for phonemes [6, 17, 69].) Sound produced without the vocal folds vibrating and only involving air flow through constrictions or past edges produced by the tongue, teeth, lips, and palate are called *unvoiced sounds*, such as for the "t" and "f" sounds.

The obstructions needed to produce consonants are formed in different *places of articulation*, including (in order from the front to the back of the mouth) *bilabial* (both lips, or labial) for the "p," "b," "m," and "w" sounds, *labiodental* (the bottom lip and upper front teeth) for the "f" and "v" sounds, *dental* or *interdental* (teeth) for both "th" sounds (as in "thin" and "them"), the alveolar ridge or *alveolar* (upper gums near the teeth) for the "d," "t," "s," "z," "n," "r," and "l" sounds, *palatal* or *alveopalatal* (hard palate, which is behind the upper gums) for the "sh," "zh" (as

in "vision"), "ch," "j," and "y" sounds, *velar* (soft palate, which is behind the hard palate) for the "k," "g," "ng," and "w" sounds, and *glottal* for the "h" sound.

The mechanical means by which consonants are formed, including the way air is pushed through the opening, is the *manner of articulation*. *Plosive* or *stop* sounds such as "d," "b," "p," "g," "t," and "k" have a staccato nature because they are formed by blocking air flow and then letting a slight rush of air. *Fricative* sounds such as the voiceless "f," "th" (as in "thin"), "s," and "sh," and the voiced "v," "th" (as in "them"), "z," and "zh" (each set in order from the front to the back of the mouth) have a hissing nature because air flow is constricted at the place of articulation, making the air flow turbulent. The "m" sound is *nasal* because the soft palate is lowered to couple the nasal cavities to the pharynx and air is suddenly released to flow through the nose. Some unvoiced sounds have a combination of initially plosive and then fricative character (and are *affricative*), such the "ch" and "j" sounds and other gutterals. The *approximants* are produced by moving one articulator to another without creating a closed constriction, such as for the "w," "y," "r," and "l" sounds. The "w" sound is also said to be formed in a *semivowel* manner.

The *acoustic signal* is the set of acoustic frequencies and intensities as a function of time. For example, Fig. 10.8a shows a sound spectrogram—a plot of the frequency components of sound versus time—for the expression "She had her dark suit in …" Distinctively different signals are seen for the different vowels and consonants. This spectrogram clearly has more useful information than the time-domain waveform for the same expression in Fig. 10.8b (even though they have the same information content).

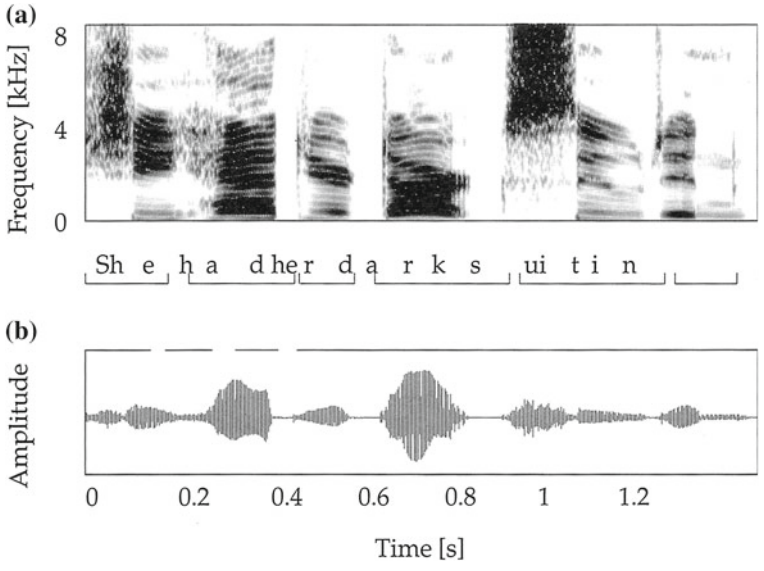


Fig. 10.8 Sound spectrogram of a female speaker saying "She had her dark suit in …" in (**a**), with the corresponding time-domain waveform signal in (**b**) (From [5])

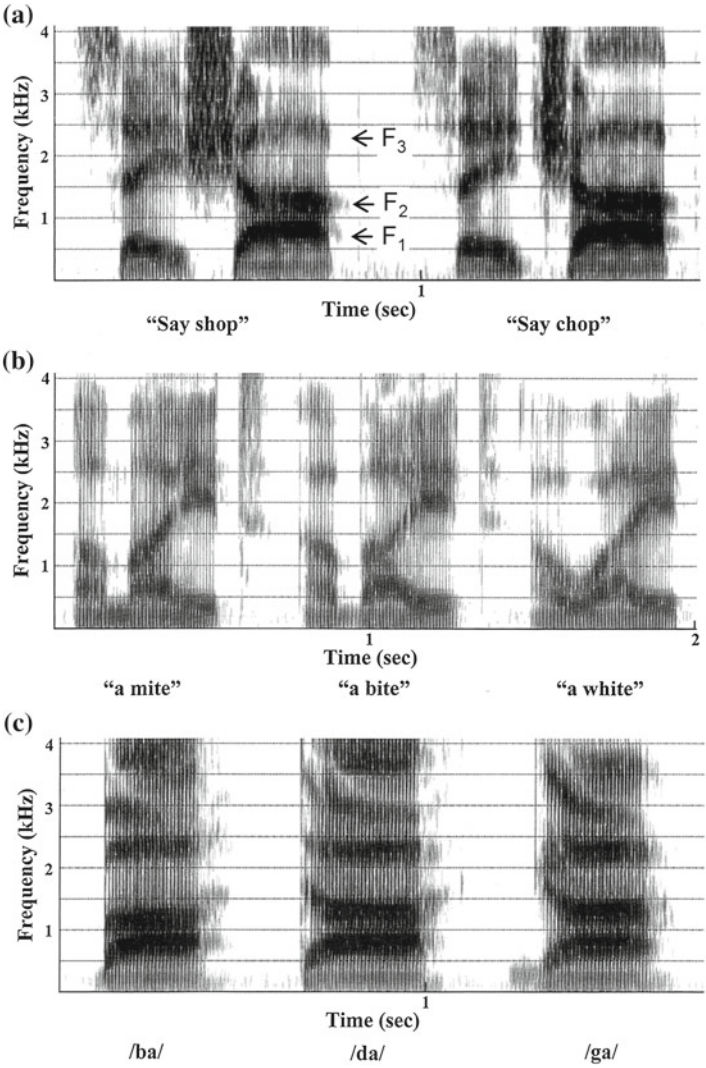


Fig. 10.9 Sound spectrograms of **a** "say shop" and "say chop," **b** "a mite," "a bite," and "a white," and **c** /ba/, /da/, and /ga/ as in "father" and "pot." The F_1, F_2, and F_3 formant frequencies for the vowel in "shop" are *denoted* in part (**a**) (From [19])

These different contributions of vowels and consonants are more clearly seen in Fig. 10.9 [5]. In part (a) the "sh" and "ch" consonant spectra are seen to be different and have different durations; also there is an interval of silence before the "ch." They are both very different from the "s" spectrum. Each vowel has characteristic bands. For example, the long a in "say" in (a) has three frequency bands, near 500, 1,700, and 2,500 Hz. These can be, respectively, labeled as F_1, F_2, and F_3. Such multiple

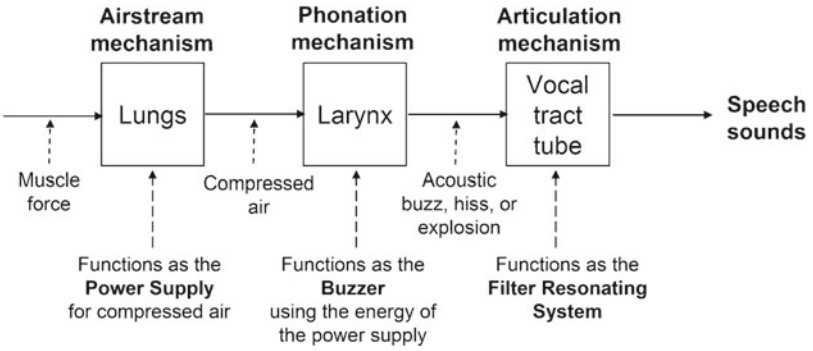


Fig. 10.10 The three neuromuscular systems in voice production (Based on [16])

bands are characteristic of vowels and are called *formants*, the first-formant (F_1), the second-formant (F_2), and so on. These formant frequencies differ for the different vowels, as is seen for this long a, the "ah" in "father" and "pot" in part (a) and /ba/ in part (c), and the long i in "mite" in part (b). As we will see later, these formants define the vowels and the characteristic feature in the perception of vowels. For all three syllables in part (c), the lowest part of the first spectrum of the consonant increases into the F_1 of the vowel. However, this variation is different for the second and third formants and this helps define these different sounds. For /ba/, F_2 and F_3 rise into the vowel; for /da/, F_2 and F_3 fall into the vowel; and for /ga/, F_2 falls, while F_3 rises into the vowel.

10.2.2 Systems in Speech Production

There are three sequential neuromuscular systems involved in speech production (Fig. 10.10). (a) The lungs are the *airstream mechanism* in which muscle force is used to produce a stream of compressed air. (b) The larynx is the *phonation mechanism*, which takes the compressed air and turns it into an acoustic buzz, hiss, or explosion. (c) The vocal tube track is the *articulation mechanism*, which takes the larynx sounds and turns them into speech sounds. These three systems, respectively, function as a power supply of compressed air, a buzzer, and a filter resonating system. We have described the properties of the lungs in Chap. 9. We will examine the phonation and articulation mechanisms now.

The Acoustic Buzzer

The separation of the vocal folds (vocal cords, glottis opening) varies with our state of speaking (Fig. 10.11). The open glottis is V-shaped because the vocal folds are held together in the front of the larynx and move apart in the back. When we are *voiceless*, the vocal folds are totally open for normal breathing. During *whispering*

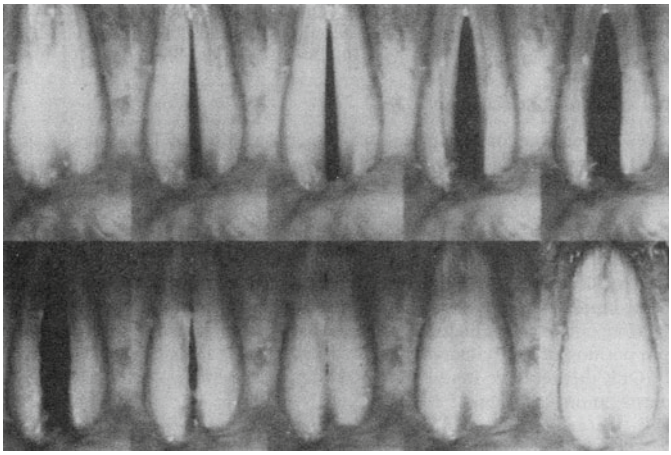


Fig. 10.11 A series of video frames of vocal-fold movement during a normal glottal *cycle*. Note that the opening is asymmetric, with the glottis more widely open at the *bottom* than at the *top*. During whispering, the glottis is even more open than during normal speaking (rightmost in *top row*), and it is even more open during forced inhalation (From [69], photo by Debra K. Klein. The University of Iowa Hospitals and Clinics. Used with permission)

the folds are closer together. Air moves through the glottal constriction (picks up speed) and rotates in turbulent eddies to give it its distinctive sound. As detailed later, our *voice* during the normal speaking of voiced sounds is created by the folds periodically opening and closing, to give periodic bursts of air. No air flows when there is a *glottal stop*.

During speaking, air rushing through vocal folds in the larynx from the trachea to the pharynx causes the vocal folds to vibrate, which in turn leads to a modulation of the air flowing into the trachea (Fig. 10.12). Initially the folds are apart and the pressure in the trachea and pharynx are equal, and both are equal to atmospheric pressure. Because of the separation, the pressure is the same before and after the folds, even with air flowing from the lungs. Muscle contractions cause the folds to move to the midline of the tube. Because of this constricted air flow, the pressure before the glottis (subglottis pressure) then exceeds the oral pressure. This pressure difference forces the folds to separate rapidly, leading to a rapid burst of air. This produces an overpressure above the glottis and an acoustic shockwave that moves up the vocal tract. The folds then rebound back to their initial positions because of their elastic recoil properties and the pressure gradient. This pressure gradient arises from the combination of Bernoulli's equation in the Venturi limit and the equation describing continuity of flow, as in (7.18):

$$P_{\text{glottis}} - P_{\text{subglottis}} = \frac{1}{2}\rho v_{\text{subglottis}}^2 \left[1 - \left(\frac{A_{\text{subglottis}}}{A_{\text{glottis}}}\right)^2\right] < 0. \tag{10.40}$$

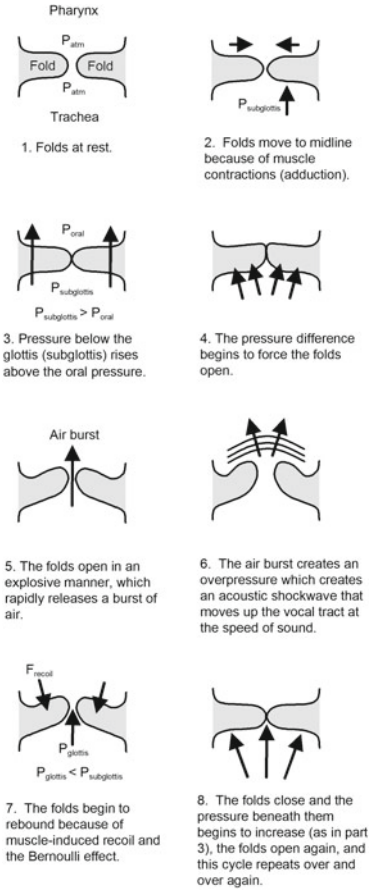


Fig. 10.12 The sequential stages of glottal vibration (Based on [16])

Because the open flow area in the glottis (between the folds) is much smaller than that below it, the flow speed in this air burst is faster and the pressure in the glottis is smaller than that beneath it. The folds return to their initial positions—with the air flow continuing. The cycle repeats and the modulated flow of air results in a buzzing sound. Figure 10.11 shows photographs of the glottis during different stages of this vibration. The motion of the glottis can be modeled with springs and other mechanical components, as shown in Fig. 10.13.

The frequency of this buzzing sound depends on the mass of the vocal folds and their tension. The fundamental frequency f_{fund} (or tone or pitch) of this buzzing is typically ~125 Hz in men and ~250 Hz in women. The lowest fundamental frequency (bass singer) is ~64 Hz (which is a low C), while the highest fundamental frequency (soprano) is ~2,048 Hz (five octaves above low C). (An *octave* extends from any given frequency to one twice that.) The acoustic buzz contains this fundamental frequency along with many harmonics (integral multiples of the fundamental).

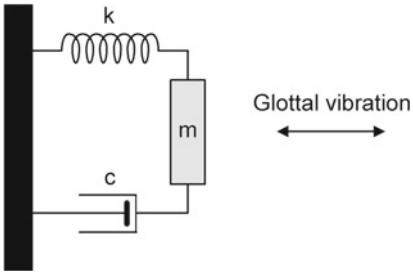


Fig. 10.13 A one-mass mechanical model of glottal vibration, with pressure against the tissue wall (Based on [69])

Each vocal fold can be modeled as a two-dimensional ribbon-like object, with vibrations along the length of the fold and in a perpendicular direction across the height of the fold. To first order, each of these vibrations can be treated with our one-dimensional models of vibration.

Using (10.37), the fundamental frequency of a string rigidly fixed at both ends, of mass density ρ, length L, and radius r under tensile force F, is $v_s/2L$, where v_s is speed of waves along the string. For such a string, we know that $v_s = \sqrt{F/\mu}$, where μ is the mass density per unit length. Using $\mu = \pi r^2 \rho$ and $v_s = \sqrt{F/(\pi r^2 \rho)} = \sqrt{[F/(\pi r^2)]/\rho} = \sqrt{\sigma/\rho}$—where the stress $\sigma = F/(\pi r^2)$—the fundamental frequency is

$$f_{\text{fund}} = \frac{1}{2L}\sqrt{\frac{\sigma}{\rho}}. \tag{10.41}$$

Most of the length of the vocal fold consists of membranous material. This length is about 16 mm for adult males and 10 mm for adult females. Figure 10.14 shows that the variation of the fundamental frequency of vocal-fold oscillation with this

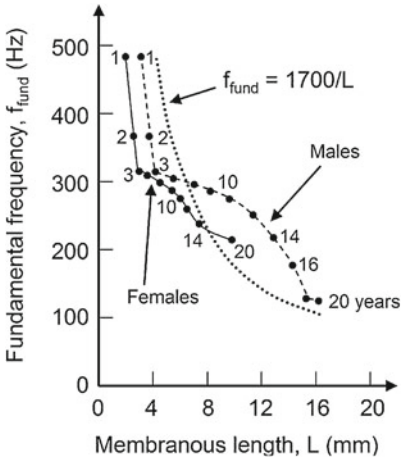


Fig. 10.14 Vocal-fold frequency as a function of length, with age as an implicit variable (Based on [69])

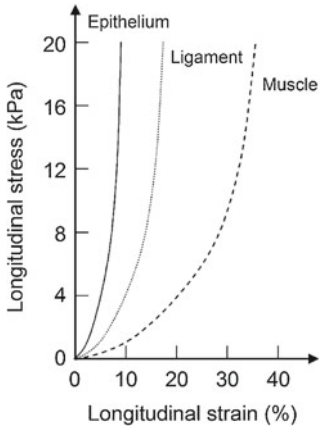


Fig. 10.15 Hypothetical stress–strain curves for tissues in human vocal folds (Based on [69])

membranous length has approximately the $1/L$ dependence predicted by (10.41). (Also see Problem 10.37.) Note this length is longer in men than women and grows with age during childhood, so the fundamental frequency is lower in men than women and decreases into adulthood. The typical stress–strain curves for human vocal-fold tissues given in Fig. 10.15 provide elastic constant information for this model (Problem 10.38). Muscular control of the folds must also be included in such models, because the oscillation is not really free.

What does the airstream leaving the glottis look like? Say that air flows into the lungs of a male during inspiration at a rate of $100\,\text{cm}^3/\text{s}$. This is also the average flow of air leaving the lungs and passing through the glottis. If the glottis vibrates at 125 Hz, a volume of $(100\,\text{cm}^3/\text{s})/125\,\text{Hz} = 0.8\,\text{cm}^3$ leaves in each air puff in a cycle. These air puffs each last for 3 ms (within the 8 ms cycle for 125 Hz), so the peak volumetric flow rate of air is $(0.8\,\text{cm}^3/3) \times 10^{-3}\,\text{s} = 270\,\text{cm}^3/\text{s}$. Because the glottis opening is typically $\sim$0.3 cm wide and it is $\sim$2 cm high, this air flows through an area of $\sim 0.6\,\text{cm}^2$. Using (7.13), the maximum speed of air leaving the glottis is $(270\,\text{cm}^3/\text{s})/(0.6\,\text{cm}^2) = 450\,\text{cm/s} = 4.5\,\text{m/s}$.

Voice-Filtering Theory (Advanced Topic)

The glottal wave (i.e., the sequence of vocal-fold pulses) from the larynx is schematically shown in the time domain in Fig. 10.16. This could represent the acoustic intensity or pressure. Let us say that it is the intensity $I(t)$. This description is in the time domain. Alternatively, we could characterize this same intensity wave using a different approach, in the frequency domain by looking at the frequencies of the sine (or cosine) waves that can be added to give the same pattern in time. The frequency components of this cyclic-glottal pattern in time can be determined by a method called Fourier analysis, to determine the strength of the fundamental I_1 (at f_{fund}) and each harmonic I_n (at nf_{fund} for $n = 2, 3, 4, \ldots$)

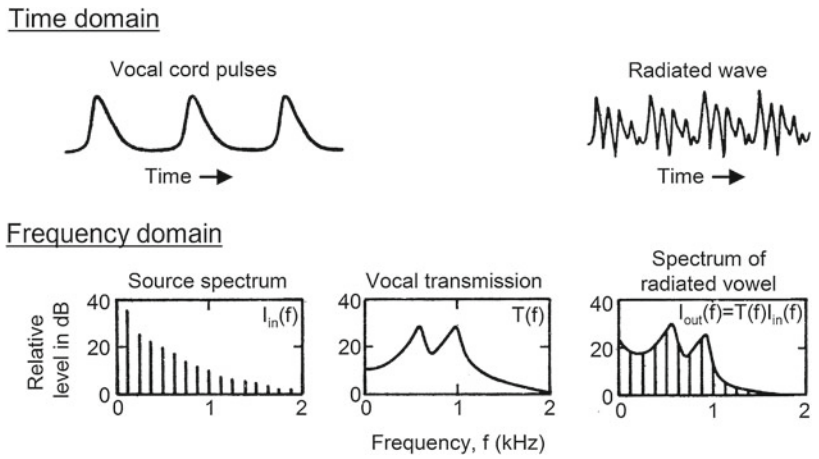


Fig. 10.16 Modification of the glottal wave by the vocal tract in the voice-filtering theory of speaking. The signals plotted versus time (in the time domain) are in the *upper row*, while those plotted versus frequency (in the frequency domain) are in the *lower row*. The source spectrum from the larynx (shown as vocal-fold pulses in the time domain) is $I_{\text{in}}(f)$, where f is frequency. The formation of the first two formants is seen in the vocal transmission ($T(f)$) and the radiated vowel ($I_{\text{out}}(f)$). The radiated vowel is shown as a radiated wave in the time domain. It is precisely our inability to conjure up a time-domain analog of the frequency-domain vocal transmission box that makes this type of frequency analysis so important (Adapted from [21]. Used with permission)

$$I(t) = \sum_{n=1 \text{ to } \infty} I_n \sin(2\pi n f_{\text{fund}} t) \tag{10.42}$$

$$= I_1 \sin(2\pi f_{\text{fund}} t) + I_2 \sin(4\pi f_{\text{fund}} t) + I_3 \sin(6\pi f_{\text{fund}} t) + \cdots . \tag{10.43}$$

(More generally, all functions of time can be *Fourier analyzed* in this manner.) This frequency spectrum associated with the glottal wave is shown in the frequency domain in Fig. 10.16, with the intensity components I_n expressed in dB SPL. (The differences in the intensity components would be more pronounced in a linear plot.) For comparison, Fig. 10.17 shows that a sine wave (or cosine wave) has a single frequency component—which is precisely the frequency of the wave, a square wave—with the same period as the sine wave—that is composed of the same frequency component plus odd harmonics of it, and a more random, aperiodic waveform in time that is composed of a wide range of frequency components.

Within the voice-filtering theory of articulation, this glottal wave (Fig. 10.16) is the input to the pharynx, and the pharynx, oral, and nasal cavities act as resonant cavities that transmit some frequencies better than others; these waves then leave our mouths and noses. We examine the frequency components of the glottal wave because it is easier to understand how this sound wave gets modified in the vocal tract by tracking each frequency component (with the transmission factor in Fig. 10.16) than by tracking the response as a function of time. The reason for this should be clear from our discussion of the simple resonant cavities formed by open and close-ended tubes.

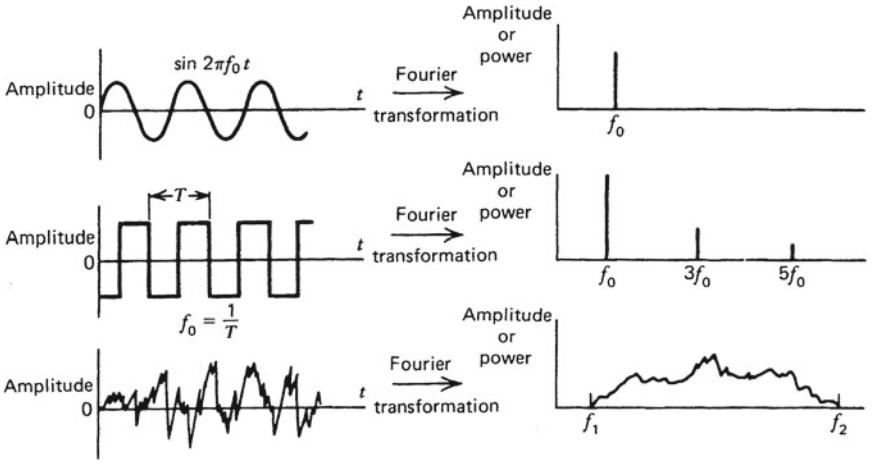


Fig. 10.17 Three waveforms on the *left* with their variations with time (the time domain), a sine (or cosine) wave, a *square wave*, and a more random pattern, with their respective frequency components (as Fourier analyzed and in the frequency domain) (From [73]. Reprinted with permission of Wiley)

In the idealized cavity of a tube, say a tube that is open on both sides, part of the sound entering one end propagates through and leaves the other end. Only those frequency components of the entering sound wave that coincide with the various resonant frequencies of the tube (the fundamental and its harmonics) will propagate through the tube and leave; more precisely, they lead to large pressures in the tube, which are then emitted by the tube. The other frequencies will be dissipated. This modification in the frequency components of the sound wave leads to a corresponding change in the sound wave in the time domain. The frequency components at each frequency f are multiplied by the transmission factors at that frequency $T(f)$

$$I_{\text{out}}(f) = T(f) I_{\text{in}}(f). \tag{10.44}$$

In reality, transmission does not occur only at precisely the resonant frequencies but in a frequency region centered about the resonant frequencies. The width of range is sometimes expressed as the frequency f divided by the Q of the cavity. These widths are relatively small in a high Q cavity. This is shown schematically in Fig. 10.7 for a generic tube. Moderate Q resonances are shown in Fig. 10.18 for several types of open-ended tubes.

The vocal tract is not a tube with these equally separated resonances, so its transmission curve is different from that of a constant diameter tube (Fig. 10.18g). The Q is quite low and so the resonances are quite broad, because of (1) the irregular shape of the vocal cavity, (2) the soft irregular nature of the walls, and (3) acoustic losses due to (a) transmission through the mouth and back through the glottis to the lungs, (b) the viscoelastic nature of the vibrating soft tissues, and (c) the viscosity of air. Because we are capable of changing the shape of the vocal tract, we

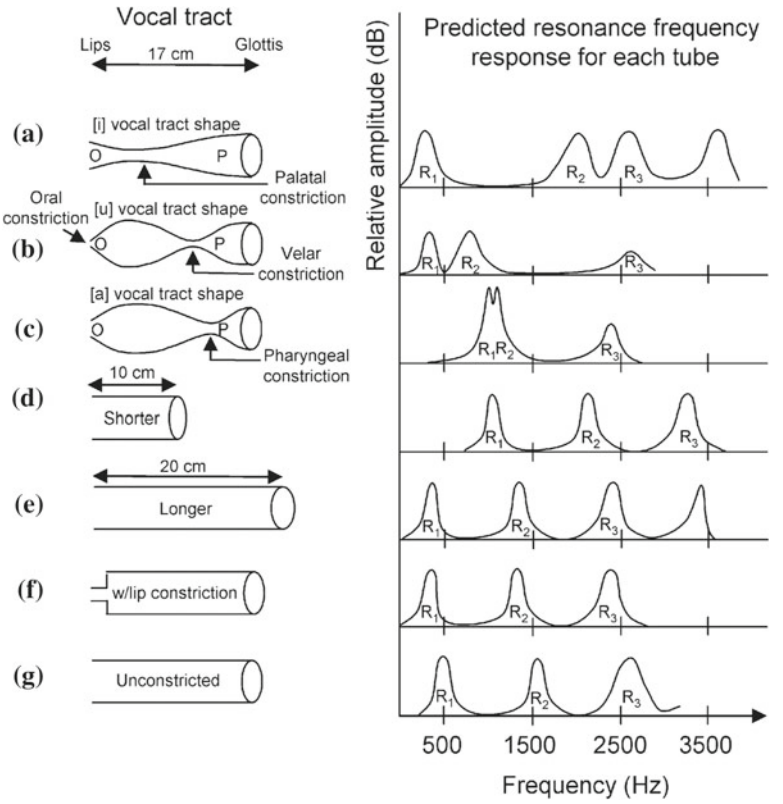


Fig. 10.18 Resonance frequencies for vocal tracts with different lengths and constrictions, relative to the unconstricted vocal track in (**g**). The vowels are **a** [i] as in "sit," **b** [u] as in "put," and **c** [a] (or "ah") as in "father" and "pot" (Based on [16])

routinely change this transmission curve during speech. For a given transmission curve $I_{\text{out},n} = T(nf_{\text{fund}})I_{\text{in},n}$ with

$$I_{\text{in}}(t) = \sum_{n=1 \text{ to } \infty} I_{\text{in},n} \sin(2\pi n f_{\text{fund}} t) \tag{10.45}$$

$$I_{\text{out}}(t) = \sum_{n=1 \text{ to } \infty} T(nf_{\text{fund}}) I_{\text{in},n} \sin(2\pi n f_{\text{fund}} t). \tag{10.46}$$

Figure 10.16 shows schematically the corresponding representations of the buzz in the time to frequency domains, the multiplication of the frequency components by the transmission factor, and the output sound signals in the frequency and time domains.

The output sound wave has the same frequency components as the input wave, including the same fundamental frequency. However, the relative strengths of each frequency component are very different. In the input wave, $I_{\text{in},n}$ decreases

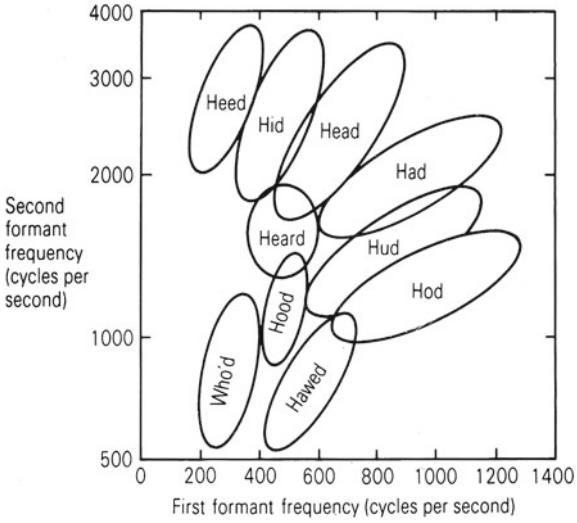


Fig. 10.19 Second-formant frequencies versus first-formant frequencies for 10 American English vowels, as spoken by a wide range of 76 people: men, women, and children. The enclosures include 90% of the sounds spoken (From [2] as adapted from [55], and used with permission)

monotonically with n (the higher the frequency the smaller the intensity). This is usually no longer true in the output wave. In fact, $I_{\text{out},n}$ peaks at several harmonics, which are called formants. The lowest frequency peak is the first formant, the second is the second formant, and so on, as is seen in Fig. 10.16.

What makes an "a" sound as in "say" different from an "e" as in "see"? They have different formants, as is illustrated for several vowels in Fig. 10.9. The actual frequencies of these formants depend on the type of sound, and they profoundly change the time variations of the sound intensity.

Figure 10.19 plots the frequency of the second formant F_2 versus that of the first formant F_1 for several vowel sounds. For each sound there is an oval showing the range of these formant frequencies for different people. For example, for the "ah" sound in "father" and "pot," the range in F_1 is 600–1,300 Hz, spanning the range from adult males to children, while for F_2 this range is 800–1,700 Hz. Even with these variations, the loci of frequencies for each sound are very different from each other (i.e., they do not overlap)—which makes it possible for us to differentiate one vowel from another spoken by anybody, even with these natural variations.

Figure 10.20 shows the acoustic spectra for soprano, alto, tenor, and bass human voices saying the vowel sound "ah" sound in "father" and "pot." In each, the fundamental and harmonics are seen to have different intensities because of the vowel formation. Evidence of the clustering of the peaks near formant frequencies is seen.

It is essential to differentiate between the different roles of the vocal-fold oscillation frequency and the vocal tract formant resonances. Figure 10.21 shows the acoustic signal in time and its frequency spectrum for the same uttered vowel with two different vocal fold fundamental frequencies. With a finer range of frequency components the short-time behavior in the time domain is smoother, but the envelope of both are similar. In contrast, Fig. 10.22 shows them for two different vowels with

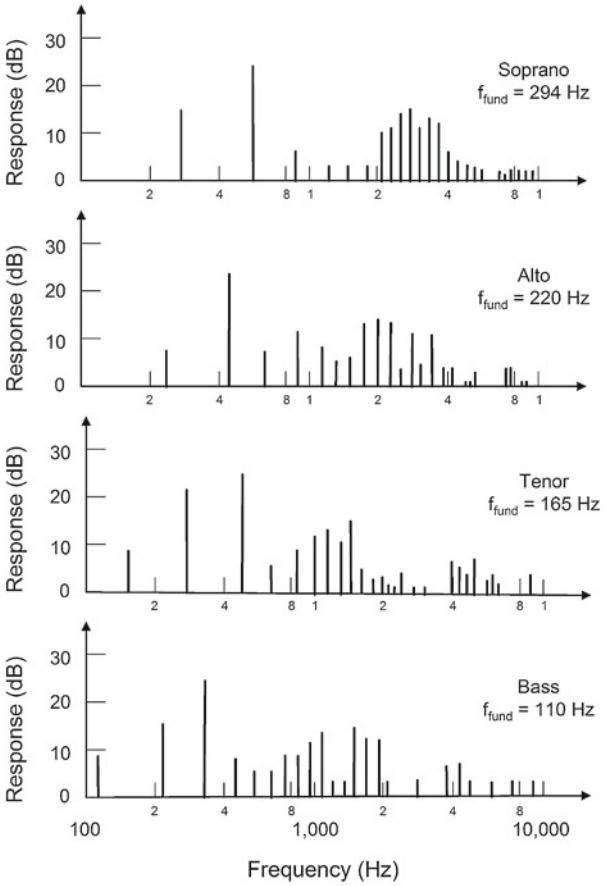


Fig. 10.20 The acoustic spectra for four voices, with the given fundamental frequencies, producing the vowel "ah" as in "father" and "pot" (Based on [54])

the same vocal fold fundamental frequency. The time domain profiles have the same short-term structure, but different envelopes. (The peaks of the envelopes are the formant frequencies.)

Figure 10.23 shows the spectral pattern of a man and woman saying the same message. In frequency space, the oscillation pattern for the man is more rapid than that for the woman, because the pattern consists of the harmonics of the fundamental buzzing frequency, which is lower in the man than the woman. However, the spectral envelope is the same for both, and this envelope carries the most important information for understanding speech.

Consider a vowel spoken by someone with a fundamental frequency of 128 Hz. If the vowel has a first formant at 512 Hz, then it is at the third harmonic of the fundamental at 128 Hz. In the time domain, this 512 Hz component is seen in the

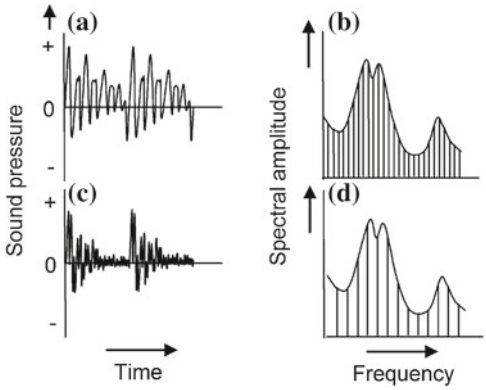


Fig. 10.21 Wave shapes (**a**, **c**) and corresponding spectra (**b**, **d**) for the same vowel "aw" as in "bought" and "awe," but with two different vocal fold frequencies of (**a**, **b**) 90 Hz and (**c**, **d**) 150 Hz (Based on [17])

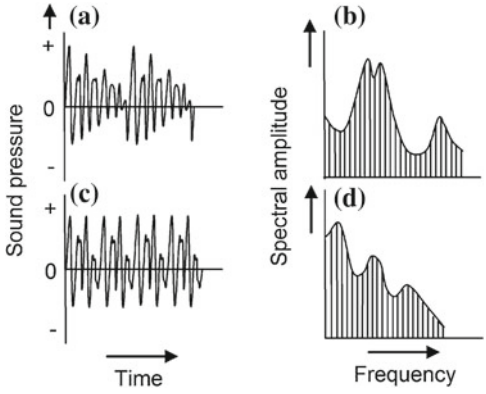


Fig. 10.22 Wave shapes (**a**, **c**) and corresponding spectra (**b**, **d**) for different vowels (**a**, **b**) "aw" as in "bought" and "awe" and (**c**, **d**) "uh" as in "but" and "about," but with the same vocal fold fundamental frequency of 90 Hz (Based on [17])

time domain signal as a fine ripple every 2.0 ms (1/(512 Hz)) superimposed on the slower repeated cycle every 7.8 ms (=1/(128 Hz)).

In the simplest model of vowel formation in the vocal tract, the region from the vocal folds to the mouth is a cylinder of constant diameter and length L that is open on one end (mouth) and closed on the other (vocal folds). (This ignores the 90° bend in the vocal tract from the pharynx to the mouth.) Using (10.39), the resonant frequencies would be: $f_m = (m + \frac{1}{2})(v_s/2L)$ which is $v_s/4L$, $3v_s/4L$, $5v_s/4L, \ldots$ for $m = 0, 1, 2, \ldots$. The formant frequencies of sound transmitted through this cavity would have ratios 1:3:5 and so on. Figure 10.24 shows that the vocal tract has very different shapes for the three depicted vowels, and consequently these vowels have very different formant frequencies. It also shows that this one-tube model gives $F_1 = 500$ Hz and $F_2 = 1{,}500$ Hz for these vowels. Calling $f_r = v_s/4L$, these correspond to f_r and $3f_r$. However, the real formant frequencies are clearly shifted from these values.

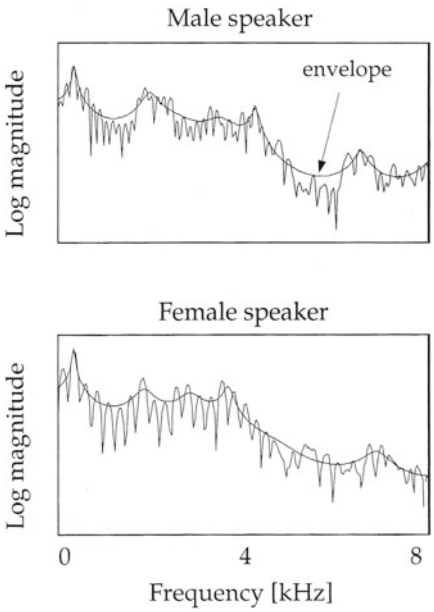


Fig. 10.23 The short-time spectra of the same linguistic pattern by a male and female speaker, with the spectral envelope shown (From [5])

Two-Tube Model (Advanced Topic)

How can we improve the one-tube model? A better model for the "ah" as in "father" and "pot," is the two-tube approximation in which the tract is modeled as a smaller diameter pharynx tube, which is closed on one end, connected to an equally long larger diameter mouth tube (Fig. 10.25a). This makes it look a bit more like the actual vocal tract in Fig. 10.24c.

Before determining the mode frequencies for this two-tube model, we need to analyze how the pressure changes when an acoustic wave travels from a tube of one diameter with cross-sectional area A_1 to one of another diameter and cross-sectional area A_2, as in Fig. 10.25a, b. This can be treated as a generalization of our earlier analysis of the transmitted and reflected acoustic beams at an interface between two semi-infinite media with different acoustic impedances. Continuity of pressure, $P_\mathrm{i} + P_\mathrm{r} = P_\mathrm{t}$, at the interface between the two tubes—as in (10.28)—is still needed; however, (10.29) needs to be modified. The motion of the media is still described by P/Z, but this quantity must be multiplied by the area A, so continuity of motion of the media requires

$$A_1 \frac{P_\mathrm{i}}{Z_1} - A_1 \frac{P_\mathrm{r}}{Z_1} = A_2 \frac{P_\mathrm{t}}{Z_2}. \tag{10.47}$$

This gives

$$\frac{P_\mathrm{r}}{P_\mathrm{i}} = \frac{Z_2/A_2 - Z_1/A_1}{Z_1/A_1 + Z_2/A_2} \tag{10.48}$$

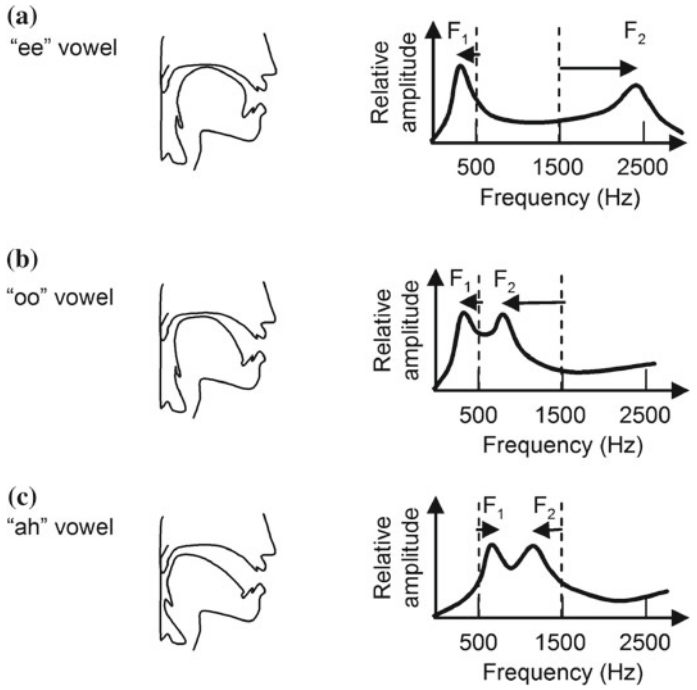


Fig. 10.24 Vocal tract shapes and spectra for three vowels: **a** "ee" as in "see," **b** "oo" as in "too," and **c** "ah" as in "father" and "pot." The difference of the first two formants are shown here relative to those for a tube with constant cross-section. These are similar to Fig. 10.18a–c, except for the more realistic vocal tract depicted here and the mouth is on the right of the larynx here (Based on [69])

and

$$\frac{P_t}{P_i} = \frac{2Z_2/A_2}{Z_1/A_1 + Z_2/A_2}. \tag{10.49}$$

Here the impedance is the same in both tubes, so

$$\frac{P_r}{P_i} = \frac{A_1 - A_2}{A_1 + A_2} \tag{10.50}$$

and

$$\frac{P_t}{P_i} = \frac{2A_1}{A_1 + A_2}. \tag{10.51}$$

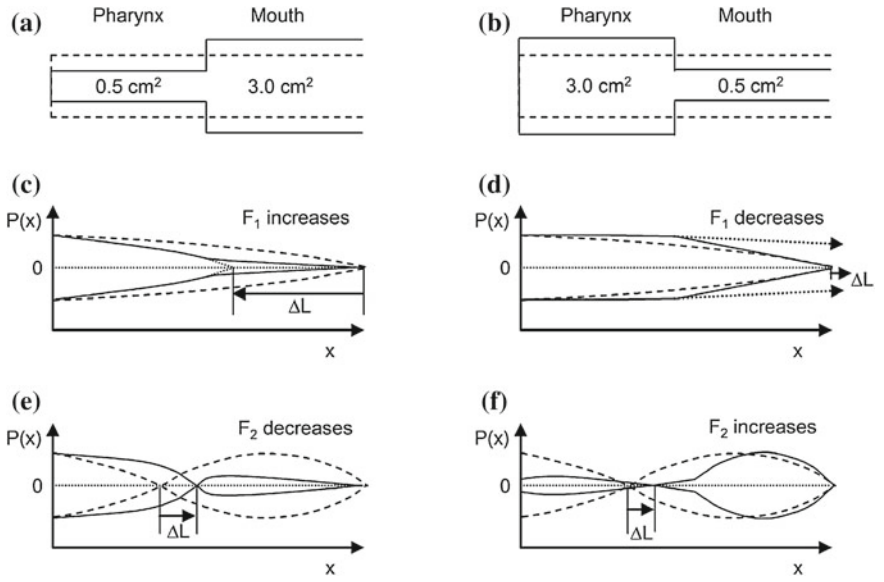


Fig. 10.25 Two-tube models of (*left side*; **a**, **c**, **e**) "ah" as in "father" and "pot," and (*right side*; **b**, **d**, **f**) "ee" as in "see," each showing the two-tube model with noted cross-sectional area in the equal length tubes in (**a**, **b**), modification of the mode for the one-tube model (*dashed lines*) to the mode in the two-tube model (*solid lines*, with changes in effective length denoted by the *dotted lines*) for the first (**c**, **d**) and second (**e**, **f**) formants (Based on [69])

This means the pressure in the transmitted wave is smaller than that in the incident wave when the second tube is wider (expansion, $A_2 > A_1$) (Fig. 10.25a) and it is greater than it when the second tube is narrower (contraction, $A_2 < A_1$) (Fig. 10.25b).

The upward shift in F_1 and downward shift in F_2 for "ah" as in "father" and "pot" relative to that of a cylinder of uniform diameter and length L, can be viewed from two perspectives. (1) The lowest two modes for a cylinder of length L change because the pressure in the wider mouth is lower than that in the narrower pharynx. This perturbs the mode structure from that of a uniform cylinder. For the lowest mode, the pressure in the mouth region is relatively less than indicated by this mode, so the opening effectively occurs (earlier in the air stream and) in the mouth, making L effectively shorter and the F_1 higher than f_r. For the next lowest mode, the node (zero gauge pressure) in the pharynx is effectively shifted to the right (toward the mouth) because the pressure should be larger in the pharynx. The length of this half wave is longer, making L effectively longer and F_2 lower than $3f_r$. This viewpoint is depicted in Fig. 10.25c, e. (2) Alternatively, we can treat the pharynx and mouth as two separate cylinders of length $L/2$ that couple weakly and therefore perturb each other. The pharynx is closed at the vocal fold end and almost open on the mouth side. Using (10.39), the lowest mode is at $v_s/(4(L/2)) = v_s/2L = 2f_r$, but since the mouth end is not totally open and there is pressure leakage to the mouth, the effective length is a bit longer, so this mode is at a somewhat lower frequency than

$2f_r$, such as the F_1 formant in this case. Furthermore, the mouth is open on the tooth end and almost closed on the pharynx end (because the pharynx is narrower). Again the lowest mode would be at $2f_r$ if the pharynx side were totally closed. Because it is not and the pressure is higher on the pharynx side, this left node is pushed into the mouth cavity and L is effectively shorter and the lowest mode is a bit higher than $2f_r$, such as the F_2 formant for this vowel. A different two-tube model seems to work for other vowels (Figs. 10.24a and 10.25b, d, f, Problem 10.45 for the "ee" sound as in "see"). For more on voice-filtering models, see [16, 21, 69].

10.2.3 Parameters of the Human Voice

Each physiological feature that helps create the voice modulates a particular feature of the vocal sound waves, and each of these acoustic features is linked to a specific type of perception. The tension in the vocal folds affects the fundamental frequency of the vocal folds—and of their harmonics, and this frequency is perceived as the *pitch* of the voice. The medial compression of the vocal folds and the pressure from the lungs affect the intensity of phonation, which leads to perceived *loudness* of the voice. The vocal-fold adduction or abduction creates an acoustic spectral difference, and this is perceived as the *quality* or timbre of the voice.

Vocal frequency. The greater the tension in the vocal folds, the greater the elastic recoil force and the thinner the folds; both effects contribute to higher oscillation frequencies. This glottal buzz rate is the fundamental frequency of the human voice. The *modal* vocal-fold frequencies, f, are those glottal buzz rates that we are most comfortable with and use most often, as in normal conversation. They typically can be varied by a factor of 2.5, with $f_{\text{highest}} = 2.5\, f_{\text{lowest}}$, by changing the vocal-fold tension; this means that people typically have a modal frequency range of 2.5 octaves. (The scale "do-re-mi-fa-so-la-ti-do" corresponds to one octave.) Even though this acoustic buzz frequency can be changed, with a concomitant change in perceived pitch, the formant frequencies do not change. (That they do not change by this factor of 2.5 should be clear from Fig. 10.19.) The modal frequency is in the lower 1/4–1/3 of the speaker's vocal range, so people typically phonate near the lower end of possible frequencies.

The average modal frequency for men is 128 Hz and that for women is 260 Hz, but there is a wide distribution about these means for men and women. Female voices are higher in pitch than male voices, because women on the average have smaller larynxes and shorter, thinner, and less massive vocal folds. The average speaking model frequencies for different voice classifications are: bass: 98 Hz, baritone: 123 Hz, tenor: 165 Hz, contralto: 175 Hz, mezzo-soprano: 196 Hz, and soprano: 247 Hz. Figure 10.20 plots the acoustic spectra for human voices in several of these categories.

Vocal intensity. The greater the medial compression of the vocal folds, the greater the subglottal pressure the closed folds can sustain before they open, and therefore the greater the loudness. Each doubling of the (gauge) pressure from the lungs increases

the voice intensity by 8–12 dB, so the vocal intensity I varies roughly as the cube of the lung pressure P_{lung}, $I \propto P_{\text{lung}}^3$. Soft speech corresponds to 2–3 cmH_2O, loud speech to 15–20 cmH_2O, and shouting to 40–60 cmH_2O. At a distance of 1 m, conventional speech corresponds to a sound intensity of about 60 dB SPL and shouting to 108 dB SPL. (Also see Fig. 10.30.)

The intensity or power of the voice in normal speech is very slowly modulated over a range of frequencies <20 Hz, for any language. This is also true for music. Screams are different in that they also have such modulations from 30–150 Hz, within the "roughness" spectral regime in this chapter. This perceptual attribute is thought to be why screams are perceived differently than normal speaking, and as alarms.

Phonation type. The vocal folds are totally open during forced inhalation and are still very open during normal breathing. They are closer together, but are still far apart during whispering and are much closer together during normal speaking—and form an obstruction to the airway. The degree that the vocal folds are open (abduction) or closed (adduction) affects the spectral distribution of the voice and therefore its quality, such as it being perceived as "noisy," "breathy" (noisy and buzzy), or "creaky" (irregularly spaced low-frequency air bursts).

The voice fundamental frequency in women decreases with advancing age and in men it decreases until 50–60 years of age, and then increases. Formant frequencies also change, but the pattern is not currently clear. The vocal folds thin and deteriorate, lose elastic and collagenous fibers, and become stiffer and thinner. These and other changes lead to some of the common attributes of the aging voice (*presbyphonia*), including increased breathiness and voice tremor, somewhat decreased volume, thinning of the voice, and lower vocal efficiency and endurance [46, 61].

10.2.4 The Energetics of Speaking

What fraction of the total energy in the air flow from the lungs during speaking is converted into the oscillatory acoustic waves that are radiated from the mouth? Say the entire vital capacity (about 5 L = 0.005 m^3) were expelled in 1 s with a lung pressure of 2 kPa (about 20 cmH_2O). This "aerodynamic" power would be 0.005 $m^3 \times 2$ kPa = 10 W during that 1 s. However, this volume could be expelled only with the glottis fully open, so with the glottis mostly closed during speaking the flow rate would be much lower than 0.005 m^3/s, and perhaps 0.0001–0.0005 m^3/s. With the same pressure differences, the "aerodynamic" power would not exceed 1 W. Therefore, continuous speaking would add only about 1% to our metabolic rate (ignoring the efficiency of forming this air flow). Conversational speech is about 60 dB SPL at 1 m, which corresponds to 10^{-6} W/m^2, and so using (10.4) the power is about $\sim 10^{-5}$ W. Defining the glottal efficiency as the ratio of this acoustic power to the aerodynamic power, the efficiency is $\sim 10^{-5} = 0.001\%$. Over a range of conditions, this efficiency is often given as ranging from 0.0001–1%.

10.3 Hearing

The ear consists of the outer, middle, and inner ears (Figs. 10.26 and 10.27). The *outer ear* is the external *auricle* (or *pinna*), which "funnels" sound into the *auditory canal* or *ear canal tube* (or *external canal* or *meatus*– mee-ay'-tus–or external auditory meatus). The *middle ear*, also known as the tympanic cavity, begins with the *tympanic membrane* (*eardrum*), and continues with the three connected small bones or ossicles, the ossicular chain: the *hammer* (*malleus*), *anvil* (*incus*), and *stirrup* (*stapes*). On one side the tympanic membrane interfaces with the ear canal tube and on the other side it is in contact with the hammer. The *inner ear* consists of the *cochlea* and the *auditory nerve* that travels from the cochlea to the brain. The stirrup is in contact with the oval window of the cochlea. The cochlea begins medially with the vestibule, continues with the anterior snail-shaped region and then in the posterior, three semicircular canals. Sound is brought into the ear by the outer ear and the actual processing of the sound begins in the inner ear. The outer and middle ear are collectively called the *conductive system*, which conducts the sound to the inner ear. We will see that the middle ear "improves" the transmission of sound from the outer ear to the inner ear. The cochlea and the eighth cranial nerve—which includes the auditory nerve as well as the vestibular nerve—are called the *sensorineural system*, which senses the sound and then creates an electrical signal and sends it to the brain. For general sources about hearing see [12, 16, 20, 22, 23, 26, 27, 29–31, 35, 36, 60, 75, 76].

There are four steps in the hearing process within the ear. (1) The sound wave enters the outer ear. (2) The movement of the tympanic membrane is transferred by conduction through the ossicles to the oval window of the cochlea. (3) The movement

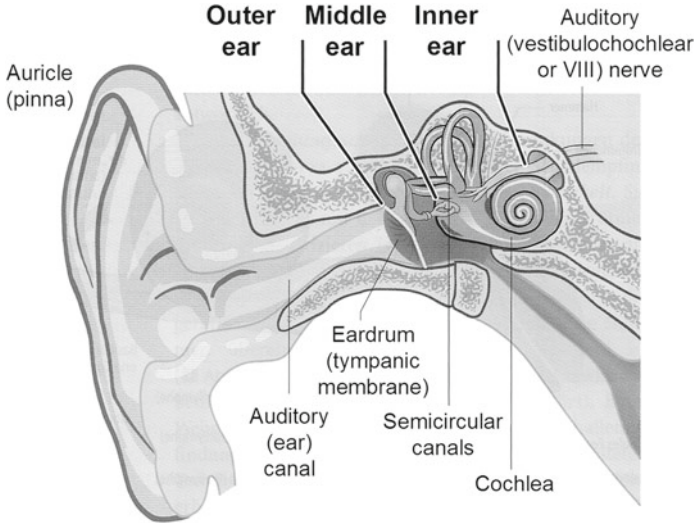


Fig. 10.26 Diagram of the outer, middle, and inner ear (From [62])

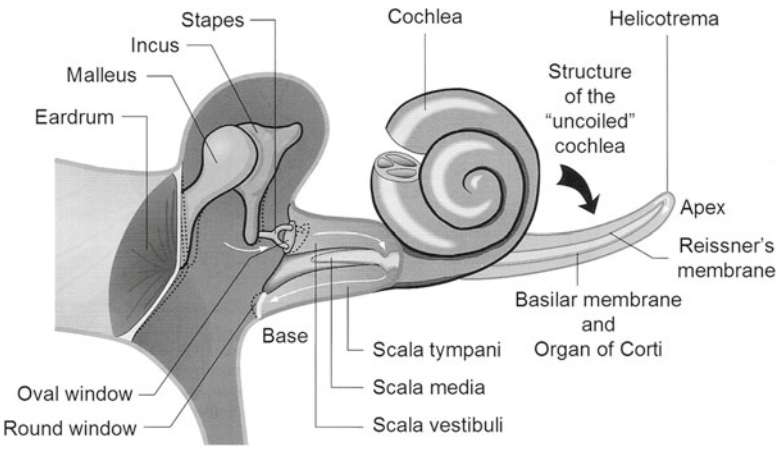


Fig. 10.27 Middle ear in detail and visualization of partial and total uncoiling of the cochlea. The cut in the partially uncoiled cochlea shows the cross-section of the tubes, which is seen in more detail in Fig. 10.32 (From [62])

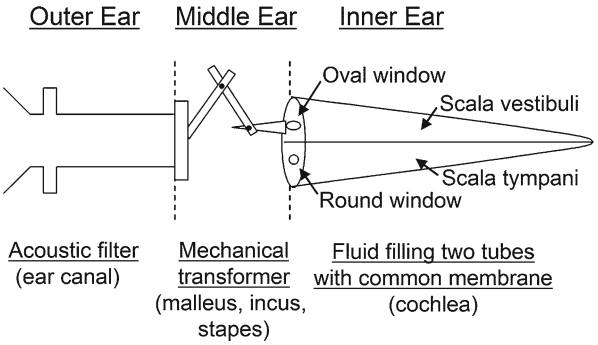


Fig. 10.28 Mechanical analogs of the outer, middle, and inner ear (Based on [30])

of the oval window generates a compressional (sound) wave in the fluid of the cochlea. (4) This wave moves the basilar membrane upon which the primary auditory receptors (the hair cells) are located, and the electrical signals generated by the hair cells are sent to the brain. This conversion of sound into electrical signals is called mechanoelectrical transduction. Mechanical model analogs of the ear are depicted in Fig. 10.28.

10.3.1 Auditory Sensitivity

The normal-human auditory system is sensitive from about 20 Hz to 20 kHz, with maximum sensitivity in a broad region near 3,000 Hz, as is seen in the audiogram of pure tone thresholds in Fig. 10.29. Between 100 and 10,000 Hz the normal threshold varies from about 0 to 20 dB SPL. The threshold for uncomfortable loudness is about

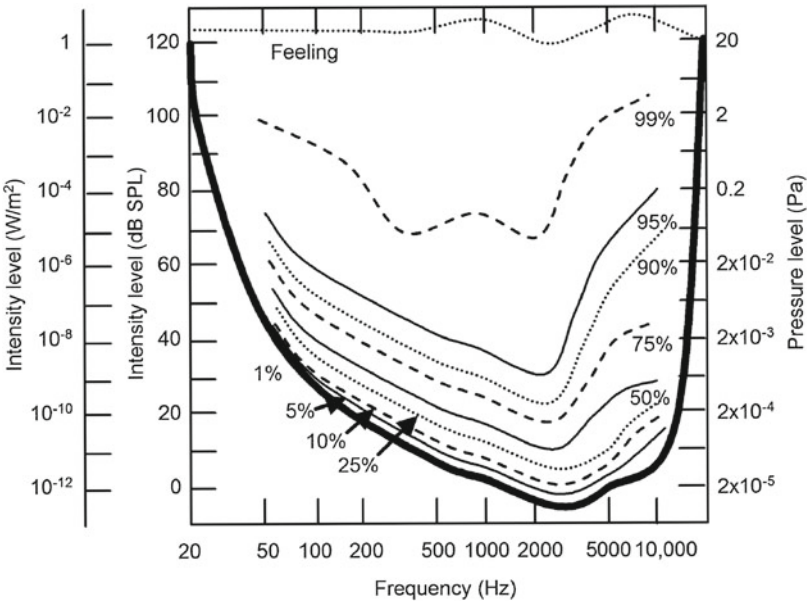


Fig. 10.29 Absolute auditory threshold for typical US residents. The *curves* show the percentage of people who could hear sounds below the level of the curve; the *top curve* shows the threshold for "feeling" in the ear (Based on [17])

100 dB SPL. From 120 to 140 dB SPL, the threshold for "feeling," "prickling," and "tickling" in the ear is reached.

Related to this is the issue of how sensitive our hearing needs to be. Our ears must be sensitive to hear our voices, and of course our voices need to be loud enough to be heard. Figure 10.30 plots the ranges of normal hearing sensitivity versus frequency, along with the ranges of intensity and frequency of speech and music, more specifically orchestral music.

Outer Ear

In spite of its funnel-like appearance, the auricle or pinna provides negligible sound collection and amplification capability. It does provide hints as to sound source localization. (Cats orient their pinnae toward a sound source to help determine its direction.) This localization depends on the asymmetrical and irregular shape of the pinna. One important feature of the pinna is that it attenuates the high-frequency components of sounds that come from the rear relative to those from the front.

The external canal in the outer ear is about 9 mm high, 6.5 mm wide, and about 2.5–3.5 cm long. It is open on one side (the outside) and terminated on the other side by the eardrum. Even though it is curved, we will model it as a straight tube, a tube open on one side and closed on the other (Fig. 10.28). With $L = 2.5$ cm, the fundamental mode has wavelength $= 4L = 10$ cm. The fundamental frequency $f = v_s/4L$ ((10.39) with $m = 0$), and with $v_s = 343$ m/s for air, we see that $f = (343\,\text{m/s})/10\,\text{cm} = 3,430$ Hz.

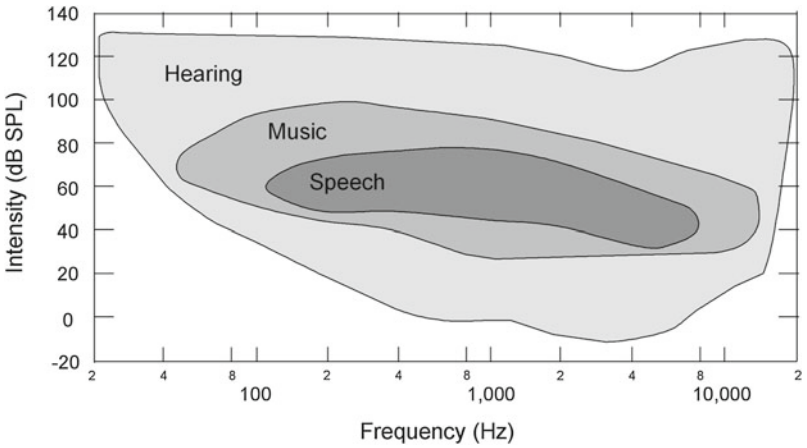


Fig. 10.30 The boundaries for normal hearing, in terms of limits of acoustic intensity and frequency, compared to the frequency and volume ranges of speech and orchestral music (Based on [10, 53])

This is very close to the peak auditory sensitivity of humans, which is near 3,000 Hz (Fig. 10.29). This makes sense because sounds near 3,400 Hz are more efficiently transported from the outer ear to the inner ear. This canal has a low Q and a rather broad resonance for several reasons, including that it is not perfectly closed.

A model of the admittance of the ear canal is presented below.

Middle Ear

Why is there a middle ear? Perhaps, it is better to ask a related question that is easier to answer: *Why does hearing become poor if the middle ear is damaged?* Without the tympanic membrane and conduction through the ossicles, sound from the external canal would be directly incident on the oval window of the cochlea and hearing sensitivity would decrease by almost 30 dB, because the oval window has a much smaller area than the tympanic membrane. This is enough of a loss to make a medium loud voice barely perceptible.

The eardrum is tilted at an angle of about 55° to the ear canal. It has an average thickness of about 0.074 mm, and is a bit taller (≈9–10 mm) than it is wide (≈8–9 mm). The malleus is ≈8–9 mm long and has a mass of 25 mg. The handle (*manubrium*) of this hammer is embedded in the eardrum by fibrous and mucous membrane layers, and is nearly upward—at 1 o'clock in the right ear and 11 o'clock in the left ear (Fig. 10.27). The incus is ≈7 mm long and has a mass of 30 mg, and looks more like a tooth with a body and two roots than an anvil. The head of the malleus connects to the body of the incus through the fairly rigid *incudomallear articulation*, which is a double saddle joint. The longer root (or process) of the incus ends in a rounded module that forms the ball in the connection to the head of the stapes (the *incudostapedial* joint). The stapes looks like a stirrup, is ≈3.5 mm long, has a mass of 3–4 mg, and has a footplate of ∼3.2 mm^2 area that attaches to the oval window by the annular ligament.

In direct conduction most of the incident sound would be reflected and very little would be transmitted for analysis by the cochlea. Acoustically, the oval window and the cochlear fluid can be approximated as being the same as water. (This is an oversimplified model. Nonetheless, cochlear fluid gives the dominant contribution to the impedance of the cochlea above 1 kHz [48].) With $Z_1 = Z_{\text{air}} = 413\,\text{kg/m}^2\text{-s}$ and $Z_2 = Z_{\text{water}} = 1.48 \times 10^6\,\text{kg/m}^2\text{-s}$ (Table 10.1), $Z_2/Z_1 = 3{,}580$ and the fraction of sound intensity transmitted into the cochlea is ((10.34) and (10.35))

$$T = \frac{4Z_2/Z_1}{(1 + Z_2/Z_1)^2} = \frac{4 \times 3{,}580}{(1 + 3{,}580)^2} = 1.1 \times 10^{-3} = 0.11\%. \tag{10.52}$$

This is a very large loss. Expressed in decibels it is a $|10 \log_{10}(I_{\text{coupled}}/I_{\text{in}})| = |10 \log_{10} 0.012| = 10 \times 2.95 = 29.5\,\text{dB}$, or about a 30 dB loss – which is very big indeed.

How does the middle ear improve this? Using jargon common in electronics, it is often claimed that the middle ear reduces the impedance mismatch by "impedance matching." This is not true. Even with the middle ear, most of the sound incident from the air on the higher impedance eardrum/ossicles/cochlear window/cochlear fluid is lost by reflection. The middle ear does improve hearing by increasing the pressure incident on the cochlear oval window via "funneling" the sound incident on the larger eardrum to the smaller cochlear window. There are several physical arguments for this (which we will present soon).

The importance of the middle ear in improving hearing is seen by the large hearing loss that results when the hammer, anvil, and stirrup are not in contact with each other in series. The *acoustic reflex* is a mechanism that intentionally and temporarily reduces this transmission through the ossicular system to limit the pressure on the oval window to protect the cochlea from very loud sounds. The stapedius and tensor tympani muscles respond in 40–80 ms as a reflex action to such loud sounds. The stapedius muscle pulls the stapes outward and the tensor tympani muscle pulls the handle of the malleus inward [33]. This mechanism also reduces low frequency sounds that often constitute background noise in a loud environment, so that information-bearing high-frequency sounds can be detected. The importance of the middle ear is also seen by one mode of permanent hearing loss, of ~30 dB, that occurs when these bones are permanently not in contact to the adjacent ossicle(s).

One mechanism of middle ear conduction is the transmission of the force on the large area eardrum (of total area ~85 mm^2, of which only 2/3 of the area vibrates, so its effective area is $A_e \sim 55\,\text{mm}^2$) to the stapes, which has a smaller area ($A_s \sim 3.2\,\text{mm}^2$) in contact with the oval window. The force on the eardrum is $F_e = P_e A_e$, where P_e is the sound pressure on it (Fig. 10.28). If this force is directly transmitted to the stapes then

$$F_e = P_e A_e = F_s = P_s A_s \tag{10.53}$$

or

$$P_s = \frac{A_e}{A_s} P_e = \frac{55 \text{ mm}^2}{3.2 \text{ mm}^2} P_e = 17 P_e. \tag{10.54}$$

With the area of the stapes on the oval window being approximately the area of the oval window, the eardrum is seen to be a much bigger sound-collection device than the smaller oval window.

This force-funneling mechanism could occur with only one ossicle connecting the eardrum and oval window. A second mechanism relies on the geometry afforded by having more than one bone in the ossicular chain. This geometry is such that the chain acts as a lever, in which torques—and not forces—are actually transmitted to the oval window. With the first lever arm being L_1 and the second L_2, torque balance gives $F_e L_1 = F_s L_2$ and $P_e A_e L_1 = P_s A_s L_2$ or

$$P_s = \frac{A_e}{A_s} \frac{L_1}{L_2} P_e = (17)(1.3) P_e = 22 P_e, \tag{10.55}$$

using $L_1/L_2 = 1.3$. Improved hearing due to this lever mechanism seems to be one reason why nature has developed an ossicular chain of bones.

These two mechanisms in the middle ear improve conduction of sound by $20 \log_{10}(P_s/P_e) = 20 \log_{10} 22 = 27$ dB. This is roughly equal to the 30 dB loss that occurs when the ossicles are not in contact with each other and that which is incurred by reflection losses from air to the cochlea; this near-equality of the middle ear improvement and the reflection losses may just be a coincidence of nature.

Other reasons for the improvement of hearing by the middle ear are sometimes cited, such as thecurvature of the tympanic membrane, which leads to the *curved membrane buckling* mechanism (Fig. 10.31). The manubrium of the malleus is attached to the center of the curved eardrum, and as a consequence of this placement it moves back and forth less than (and about half as much as) the average back and forth displacement of the eardrum. The argument continues that the force on the manubrium is double that of the membrane because the same torque is transmitted from the eardrum to the malleus and the manubrium is moving half as much as the

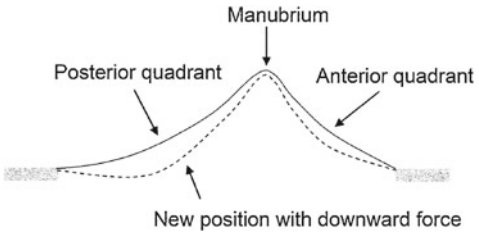


Fig. 10.31 Curved membrane buckling principle for the eardrum, with the position of the manubrium of the malleus shown. The membrane is shown without a force (*unbroken lines*) and with a (*downward*) force (*dashed lines*) (Based on [27, 70])

membrane. Some think the curved membrane buckling effect doubles the overall middle ear mechanical advantage from 22 to about 44; if it does, this leads to a total increase of 33 dB due to the middle ear.

We will now analyze the vibration of an eardrum under tension, which is a vibration that is normal to a 2D surface, by modeling it as the vibration of a string under tension, which is a vibration normal to a 1D object. Using (10.41), the fundamental frequency of a string under tensile stress σ that is rigidly fixed at both ends is $(1/2L)\sqrt{\sigma/\rho}$, for mass density ρ and length L. For a circular membrane of radius a, we substitute $2a$ for L, to get $(1/4a)\sqrt{\sigma/\rho}$. Thex fundamental frequency of the transverse vibration of a circular membrane of radius a, thickness d, and mass density ρ under tension T is actually

$$f_{\text{fund}} = \frac{2.405}{2\pi a}\sqrt{\frac{\sigma}{\rho}}, \tag{10.56}$$

where the stress $\sigma = T/d$. (The derivation of this exact result is beyond our scope.) The agreement with the simple string model is surprisingly good. Consider an eardrum with a diameter of 8 mm, a thickness of 20 μm, and mass density $\rho = 1{,}000\,\text{kg/m}^3$. For it to have a fundamental resonance at 1,000 Hz, the stress must be $1.1 \times 10^5\,\text{N/m}^2$ and the tension must be 2.2 N/m. One would expect the eardrum to have a resonant frequency roughly near the 3,000 Hz peak auditory sensitivity and resonance of the ear canal.

Inner Ear

The vibration transmitted to the stapes is transmitted to the fluid in the cochlea by the stapes being in contact with the oval window. The vibration in the cochlear fluid induces motion which produces an electrical signal that causes the release of a chemical transmitter (Chap. 12). This transmitter excites nerve cells that are connected to the brain. Models of the cochlea are described in [3, 12, 15, 41, 64]. (Much ground-breaking work on the cochlea was performed by Georg von Békésy, who was awarded the Nobel Prize in Physiology or Medicine in 1961 for his discoveries of the physical mechanism of stimulation within the cochlea.)

Cochlear Tube Vibrations

The cochlea can be viewed as a tube rolled into $2\frac{3}{4}$ turns. Uncoiled, as shown in Fig. 10.27, it would be a cylinder ∼2 mm in diameter and 35 mm long. It has an "upper half" tube, the *scala vestibuli*, and a "lower half" tube, the *scala tympani*, that contain a fluid called *perilymph* and are separated by a cochlear partition consisting of the *scala media* and the *organ of Corti*, which contain a fluid called *endolymph* (Fig. 10.32). The scala vestibuli and scala tympani flow into each other at the *helicotrema*, which is at the end of the tube at the apex. The organ of Corti contains the hair cells and abuts the scala tympani. Sound is transmitted by the stapes into the scala tympani through the oval window and the coupling between the scala tympani and scala vestibuli sets the *basilar membrane* (between the scala tympani and the hair

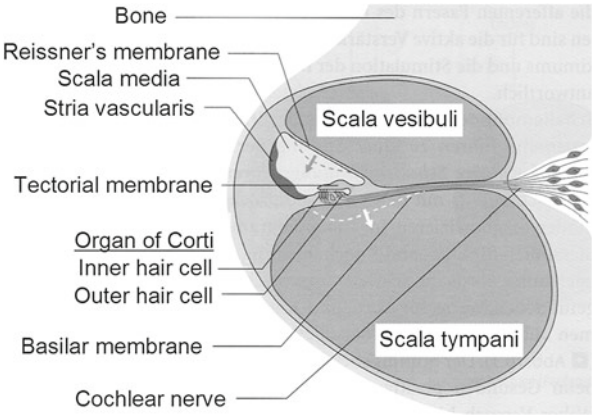


Fig. 10.32 Cochlea cross-section. (The cochlear nerve combines with nerves from the semicircular canals to form the auditory nerve.) (From [62])

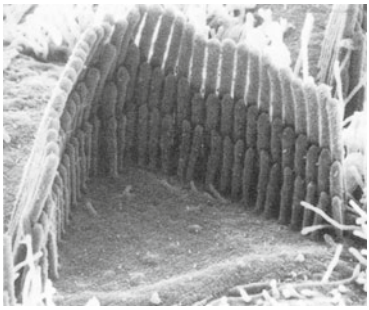


Fig. 10.33 Scanning electron micrograph of a single outer hair cell bundle from a mammalian cochlea (From [34]. Used with permission)

cells) moving transverse to the membrane, thereby exciting the hair cells (Fig. 10.33). This motion is like the traveling wave set in motion by the snapping of a rope.

There are 3,500 flask-shaped *inner hair cells* and 12,000 tube-like *outer hair cells*, which physically differ in how the hair in the cells protrude [22]. The inner hair cells relay information about the sounds transmitted to the cochlea to the brain. The outer hair cells increase the amplitude and frequency selectivity of the vibration of the basilar membrane for low-level sounds. About 95% of the auditory nerve cells receive their signals from the (relatively fewer) inner hair cells because each inner hair cell connects to about 8–30 auditory nerve fibers, while several outer hair cells connect to the same nerve fiber (Fig. 10.34). When the hair bends in one direction it depolarizes and in the opposing direction it hyperpolarizes, and this alternation causes the hair cell to release the neurotransmitter. (See Chap. 12 for a description of polarization.) Motions as small as 10^{-10} m can generate a response in a hair cell.

The basilar membrane becomes 3–4× wider and a hundred-fold less stiff along the "uncoiled" cochlea from its beginning near the stapes (the base) to its end (the apex) (Fig. 10.35). Both of these characteristics help resolve the frequency components of the sound wave in such as way that the higher frequency components have larger amplitude vibrations nearer the stapes and the base, while the lower frequency components have larger amplitude vibrations nearer the apex (Figs. 10.36 and 10.37).

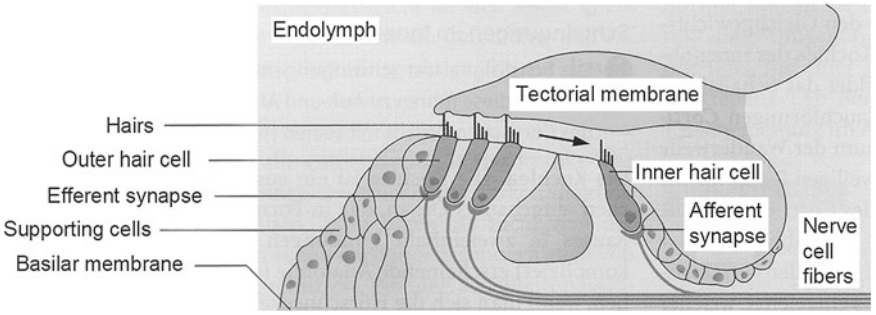


Fig. 10.34 Inner and outer hair cells in Organ of Corti (From [62])

Fig. 10.35 The uncoiled cochlea gets narrower from the base at the stapes to the apex, as the basilar membrane gets wider (Based on [29, 63])

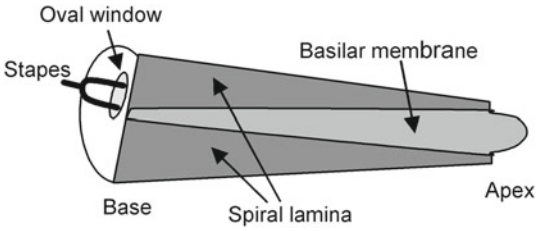
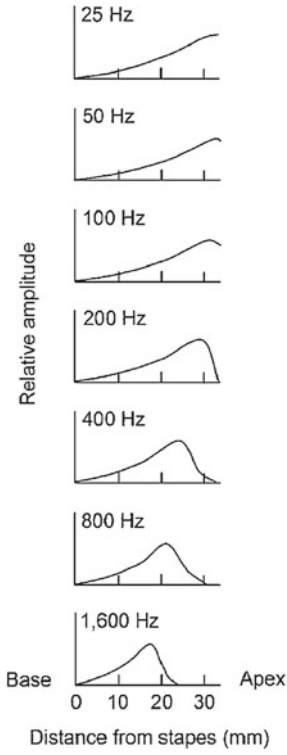


Fig. 10.36 Envelope of basilar membrane vibration versus frequency (Based on [9, 29])



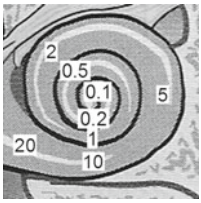


Fig. 10.37 Tonotopic map of cochlear sensitivity to audio frequencies (shown in kHz) (From [62], and based on [13, 22, 29])

This spatial separation of the frequency components means that the hair-cell-excited neurons progressively farther down the cochlea are selectively excited by progressively lower frequency components, and these frequency analyzed signals are sent as such to the brain. This frequency analysis is improved further by the changing characteristics of the hair cells along the cochlea, because the resonant frequencies of the hair cells are tuned to match the changing optimized local frequency. This is called a *place code* for frequency. There is also a *time code*: the higher the acoustic frequency, the higher the neural firing rate. Because the maximum firing rate of these neurons, 500 Hz, is too slow for a single neuron to fire at most moderate to high acoustic frequencies, several neurons cooperatively fire. For instance, for a 3,000 Hz wave, this *volley principle* suggests that six neurons could fire successively, with say neuron #1 firing at cycles 1, 7, 13, . . ., neuron #2 firing at cycles 2, 8, 14, . . . and so on.

The higher the acoustic intensity, the faster, in general, the rate auditory neurons fire. However the dynamic range of response is only about 40 dB, and most fibers reach saturation above 40–50 dB SPL and do not fire at faster rates at higher intensities. Higher intensities are recognized as such (possibly) because some neurons do not begin to fire until the mid-intensity range or some neurons are located in places where they do not fire much until the intensity is high.

We could try to calculate the resonance modes of the basilar membrane by analyzing the waves in a structure that looks like an uncoiled cochlea. One simplified model is a tapered circular-sector membrane of uniform stiffness, as is shown in Fig. 10.38. Obtaining the analytical solution for the vibrations of this structure is beyond the level of this text; it is plotted in this figure. (Also see [53, 54].) The waves in this model semiquantitatively agree with the data in Fig. 10.36, the lower the frequency of the wave, the closer its maximum is to the apex. Inclusion of the stiffness variation in the model requires a numerical solution, and this is expected to accentuate the spatial separation even more.

The outer hair cells respond to vibration much as the inner hair cells, but contribute little to the signals sent to the brain. These cells become shorter when they are depolarized and longer when they are hyperpolarized, by as much as 4% when they are isolated. This *cochlear amplifier* mechanism may help magnify the basilar membrane motion and improve the frequency selectivity of the auditory detection process, as is illustrated in Fig. 10.39.

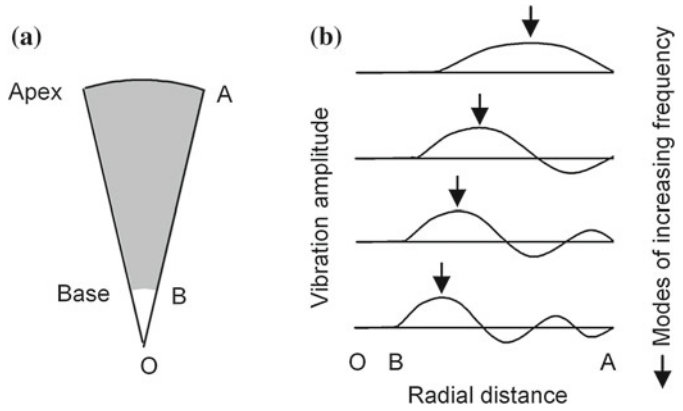


Fig. 10.38 Mechanical vibration model of the uncoiled cochlea in Fig. 10.35 modeled as a tapered membrane with uniform stiffness in (**a**), with predictions of the first few modes in (**b**). The principal maxima (at the positions of the *arrows*) of the modes move to the base with increasing frequency, which agrees with the trend in Fig. 10.36. For an untapered membrane, the modes would look the same except each antinode (maximum and minimum) in the mode would have the same magnitude (Based on [23])

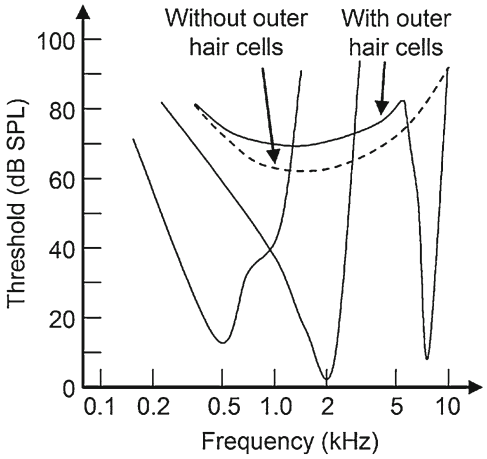


Fig. 10.39 The excitation threshold for nerve hair cells or auditory nerve fibers with three different resonant or characteristic frequencies. Without outer hair cells, the sound threshold is much higher and the frequency selectivity is much poorer, as is illustrated for the fiber with the highest of the three resonant frequencies (Based on [22])

Hair Cell Vibrations

We will first try to model the transverse vibrations of hair fibers of length L (in hair cells) as the acoustic longitudinal vibrations of a tube that is closed at one end (into the body of the cell) and open on the other (to sense the vibrating fluid). (This is also analogous to the transverse vibration of a string that is fixed on one end and free on the other.) Using (10.38) and (10.39), the resonant wavelengths are $4L$, $4L/3$, $4L/5$, ... and the frequencies are $v_s/4L$, $3v_s/4L$, $5v_s/4L$, ... With $v_s = \sqrt{Y/\rho}$ the lowest frequency is $v_s/4L = (\sqrt{Y/\rho})/4L$. However, this is really the resonant frequency

for longitudinal vibrations across a wide bar of length L, with one face fixed and the other free.

Transverse oscillations of a bar are quite different [53, 54]. For a cylinder of length L and radius a, the lowest resonant transverse frequency is

$$f_{\text{fund}} = \frac{\pi a \beta^2}{4L^2}\sqrt{\frac{Y}{\rho}}, \tag{10.57}$$

where $\beta = 0.597$. (The derivation of this is beyond our scope.) This assumes the bar is attached to a base with no compliance. As the base becomes more compliant, the value of β decreases, down to $\sim$0.35 for equally compliant bases and rods, down to $\sim$0.2 for bases that are 10-fold more compliant than the rod, and so on. Assuming hair is composed of keratin with $Y = 2 \times 10^9$ Pa and density $\rho = 1{,}000\,\text{kg/m}^3$, the lowest resonant frequency of a hair of length 2 mm and diameter 20 μm that is firmly mounted in an incompliant base is 1.0 kHz. (The cell is likely a bit compliant, so its resonant frequency is likely a bit lower.)

Admittedly, this model is perhaps too simplified. Several features should be added to it, including the viscosity of the cochlear fluid and physical connections between the hair cells. Moreover, it is not clear if there is a connection between hair cell vibration frequencies and auditory frequency selection in mammals.

The Immittance of the Human Ear (Advanced Topic)

We have analyzed the mechanical behavior of several components of the outer, middle, and inner ears. *How do these and other components combine to determine the acoustic immittance (impedance, Z, or admittance, $Y = 1/Z$) of the ear?* The stiffness (the stiffness (or negative) reactance, X_s or, equivalently, the stiffness (compliant) susceptance, B_s) comes from the air in the middle and outer ear spaces, the eardrum, and the tendons and ligaments in the ossicular chain. The inertial mass terms (the mass (positive) reactance, X_m, or the mass susceptance, B_m) come from the ossicles, the pars flaccida part of the eardrum, and the fluid in the scala vestibuli and scala tympani in the cochlea—the perilymph. The resistive, frictional component (the resistance, R, or the conductance, G) is produced by the perilymph, the mucous membrane linings of the middle ear, the narrow passages between the middle ear and mastoid air cavities, the eardrum, and the tendons and ligaments in the middle ear, and is mainly due to the absorption of the energy of the traveling wave by the cochlea.

It is easier to work with admittances here than with impedances because they are additive. (Why?) The admittance is measured with a probe that sends a 85 dB SPL signal, usually at 226 or 220 Hz, into the outer ear, and monitors the reflected signal with a microphone [27, 50]. This is all performed as a function of air pressure in the ear canal, with this (gauge) static (and not sound) pressure varied in time from about +200 daPa to -300 daPa. (The unit dekapascals (daPa) is common in audiology, with 1 daPa = 10 Pa = 0.075 mmHg = 1.02 mmH_2O.) Middle ear pathologies and middle ear muscle contraction (the acoustic reflex) can be analyzed with these

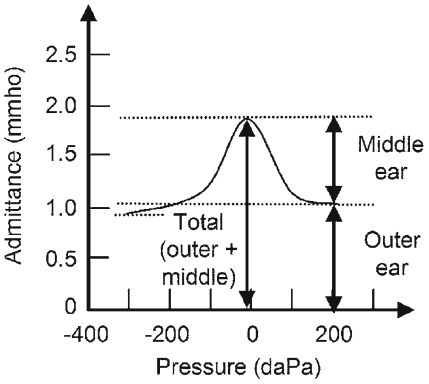


Fig. 10.40 Typical admittance tympanogram at 226 Hz, showing the different contributions to the acoustic admittance. Note that the outer ear admittance determined at −300 and −400 daPa is lower than that determined at +200 daPa, so the middle ear admittance determined at −300 and −400 daPa would be larger than that at +200 daPa (Based on [27])

tympanograms. The middle ear admittance, $Y_{\text{middle ear}}$, is determined by measuring the total admittance of the outer and middle ears at ambient pressure (the eardrum is relatively acoustically transparent then) to get $Y_{\text{total}} = Y_{\text{outer ear}} + Y_{\text{middle ear}}$, and subtracting from that the admittance of the outer ear only. $Y_{\text{outer ear}}$ is measured by pressurizing the outer ear, because the pressurized eardrum is acoustically opaque to the middle ear. From Fig. 10.40, $Y_{\text{outer ear}} = 1.0$ mmho (its value at 200 daPa) and $Y_{\text{total}} = 1.75$ mmho at 0 daPa, and we find $Y_{\text{middle ear}} = 0.75$ mmho. (At −300 daPa the admittance is actually lower than at 200 daPa, and it may provide a better value of $Y_{\text{outer ear}}$ than at 200 daPa.)

The ear canal volume expressed in cm^3 is the same number as the outer ear admittance expressed in mmho, when it is measured at 226 Hz. This means that $Y_{\text{outer ear}} = 1.0$ mmho, which indicates that the ear canal volume is $1.0\,\text{cm}^3$ (= 1.0 mL). Is this a coincidence or something that is deep? We will now see that this is in part a coincidence of the natural units and in part caused by choosing $f = 226$ Hz.

A Model of Air Vibrating in the Ear Canal (Advanced Topic)

How can we determine the admittance of the outer ear Y_{outerear}*?* Let us model the ear canal as a closed tube of volume $V = AL$, where A is its (constant) cross-section and L is its length [32, 53, 54]. Initially, the internal and external pressures are the same, P. If this tube is squeezed, the pressure inside increases by an amount ΔP and this will lead to a restoring force $(\Delta P)A$, like the restoring force of any spring-like material.

The ideal gas law (7.2) is $P = nRT$, where P is the pressure, n is the density ($= N/V$, where N is the total number of molecules in the volume V), R is the gas constant, and T is the temperature. If the volume were distorted to $V + \Delta V$, with the number of molecules kept constant and the temperature also constant (which is called an *isothermal* condition), the pressure would change to $P + \Delta P$ with PV and $(P + \Delta P)(V + \Delta V)$ both equal to NRT and to each other. So

$$PV = (P + \Delta P)(V + \Delta V) \simeq PV + (\Delta P)V + P(\Delta V), \tag{10.58}$$

where the term $(\Delta P)(\Delta V)$ is small and has been neglected. This gives

$$\Delta P = -\frac{P(\Delta V)}{V} = -\frac{P(\Delta L)}{L} = -\frac{nRT(\Delta L)}{L} \tag{10.59}$$

using the change in volume, $\Delta V = A(\Delta L)$, assuming area A is constant in the distortion and $P = nRT$.

It is conventional to normalize all terms by A^2, including the driving force $(\Delta P)A$, so

$$\frac{(\Delta P)A}{A^2} = \frac{\Delta P}{A} = -\frac{nRT}{V}(\Delta L) = -\frac{nRT}{V}x = -\frac{P}{V}x, \tag{10.60}$$

where now ΔL is called x. The coefficient $nRT/V = P/V$ is the spring constant k in (10.21), which was also called S in the earlier analysis.

We have assumed that the temperature does not change during the squeezing. If we instead assumed the more realistic condition that heat flow across the tube walls can be neglected during the squeezing and response (which is called an *adiabatic* condition), then

$$k = \frac{\gamma nRT}{V} = \frac{\gamma P}{V}, \tag{10.61}$$

where γ is the parameter in (10.2). (This is shown in Problem 10.59.) With $k = S$ and $X_s = S/\omega = S/2\pi f$

$$X_s = \frac{\gamma P}{\omega V} = \frac{\gamma P}{2\pi f V}. \tag{10.62}$$

When this is the main term in the magnitude of the impedance, then $| Z | = X_s$ and the magnitude of the admittance is then $| Y | = B_s = 1/X_s$, giving

$$| Y | = B_s = \frac{2\pi f}{\gamma P} V. \tag{10.63}$$

An ambient pressure P of 1 atm. = 0.1 N/mm^2 = 1×10^6 g/cm-s^2. For air $\gamma = 1.4$ and so at 226 Hz we find that $2\pi f/\gamma P = (2\pi)(226\,\text{Hz})/(1.4(1 \times 10^6\,\text{g/cm-s}^2)) = 1.0 \times 10^{-3}$ cm-s/g. Because 1 mho = 1.0 cm^4-s/g and 1 mmho = 1.0×10^{-3} cm^4-s/g, we see that a volume $V = 1.0\,\text{cm}^3$ has an admittance of 1.0 mmho. Using the scaling arguments developed in Chap. 8 (8.11), the admittance of the outer ear expressed in mmho gives the volume of the ear canal expressed in cm^3.

Hearing Loss

Sound levels above 85 dB SPL are considered harmful, those above the pain threshold of 120 dB SPL are unsafe, and those above 150 dB SPL cause physical damage to the human body. To avoid hearing damage, safety organizations recommend exposure



Fig. 10.41 Potential sources of hearing loss. (For sure, the current reader of this book does not share the complaint of the "dude" on the *right*. Still, we should interpret the interaction between the dude on the *right* and his mother from a physics perspective. His mother wants him to decrease the entropy (or disorder) of his room. He has (unstated) concerns about this because he will need to use energy to decrease the room entropy and he knows that the overall entropy of the universe would increase as a result of any effort on his part to clean up his room) (From Beattie; Daytona Beach News-Journal, Copley News Service, NJ-CENTER.com, copyright 1999. Reprinted with permission)

to no more than 85–90 dB SPL for 8 h a day, 100 dB SPL for 2 h a day, or 110 dB SPL for 30 min a day. Eardrums rupture at 190 dB SPL to 198 dB SPL and sound levels of around 200 dB SPL can cause death to humans. Such levels are generated near bomb explosions, such as 3 m away from the detonation of 23 kg of TNT. Other potential causes of hearing loss are suggested in the cartoon in Fig. 10.41.

The main way hearing loss is evaluated is by determining the threshold for hearing pure tones from about 125–8,000 Hz. The hearing level is usually expressed as the threshold value of hearing expressed in dB HL versus audio frequency, with the clinical plot showing increasing threshold intensities toward lower ordinate values (Fig. 10.42b), rather than toward higher ordinate values, as in Fig. 10.29 or 10.42a. HL stands for "hearing level." At each frequency, the threshold expressed in dB HL is that in dB SPL minus the reference normal value in dB SPL. This accounts for the variation in threshold intensity with frequency that is seen in Figs. 10.29 and 10.42a. These reference values are about 7.5 dB SPL from 750 to 1,500 Hz. At lower frequencies they increase to $\simeq$13 dB at 500 Hz, 26 dB SPL at 250 Hz, and 47 dB SPL at 125 Hz. At higher frequencies they increase to $\simeq$10.5 dB SPL from 2,000 to 4,000 Hz and 13.5 dB SPL from 6,000 to 8,000 Hz.

Normal hearing is therefore defined as 0 dB HL at all "audible" frequencies (top, flat curve in Fig. 10.42b), but people with an average frequency response $\leq$15 dB HL

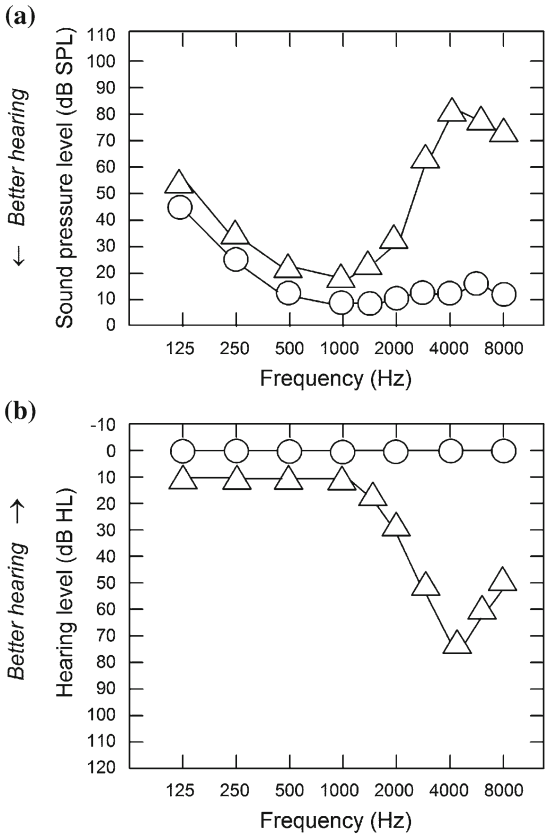


Fig. 10.42 Audiograms in **a** dB SPL and **b** dB HL units for a normal person (*circles*) and one with high-frequency hearing loss (*triangles*) (Based on [26, 27])

are still said to have "normal hearing." Hearing loss is defined as slight (average value of 16–25 dB HL), mild (26–40 dB HL), moderate (41–55 dB HL), moderately severe (56–70 dB HL), severe (71–90 dB HL), and profound (>90 dB HL). An example of a real audiogram for a person with normal hearing is shown in Fig. 10.43. Figure 10.44 shows how hearing sensitivity decreases with age, which is called *presbycusis*. One such example is plotted in dB HL in Fig. 10.45.

Modes of hearing loss can be caused by problems in the middle ear or the inner ear. The regular route of receiving sound from the outer ear to the inner ear is called *air-conduction*. Sound can also be conducted to the inner ear by *bone-conduction*, in which sound is transmitted by vibrations of the bones of the skull. Both induce the same cochlear activity. In clinical audiology, air-conduction is tested by using earphones or loudspeakers, while bone-conduction is tested by a vibrator placed on the front of the skull. The modes of skull vibration change with vibrator frequency, as is seen in Fig. 10.46. At 200 Hz the skull vibrates in unison with the same phase in the forward and backward directions. When the frequency is increased to 800 Hz, the front and the back of the skull are out of phase with each other, and at 1,600 Hz there

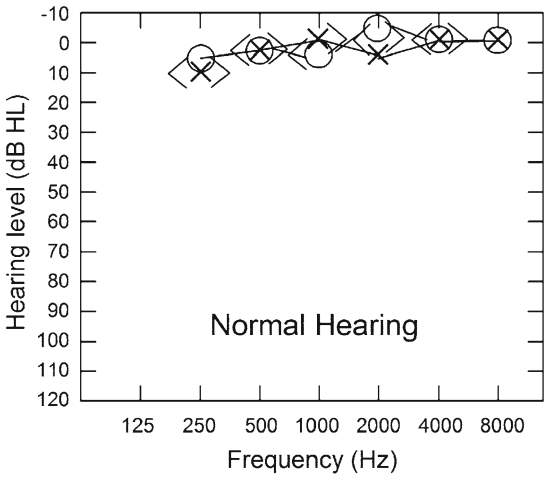


Fig. 10.43 Example of a real audiogram for a person with normal hearing. Audiogram key: right air-conduction (*open circles*), left air-conduction (*cross symbols*), right bone-conduction (*left angular brackets*), and left bone-conduction (*right angular brackets*), all unmasked (Based on [27])

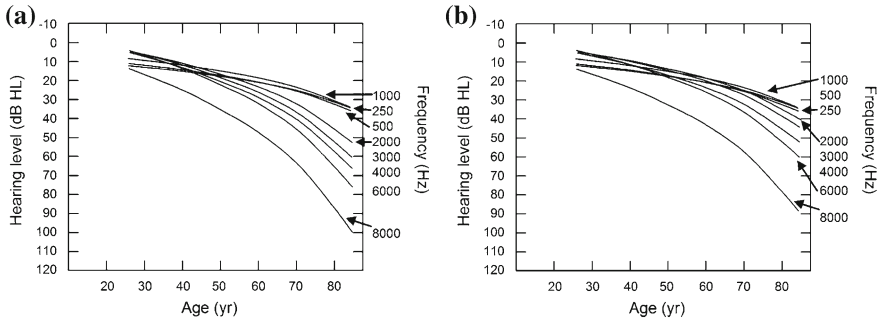


Fig. 10.44 Hearing levels versus age for **a** males and **b** females (Based on [27, 42, 65]. Also see [59] for more data)

are forward–backward and right–left vibrations. Above ∼1,000 Hz, skull vibrations lead to important hearing response by excitations of the outer, middle, and inner ear; at lower frequencies the middle and outer ear mechanisms are most important.

If someone has poor hearing due to a problem in the middle ear (or excessive wax in the ear canal), sound can still propagate through the bones in the head to the inner ear. Thresholds for conduction to the ear by air and by direct transmission to bone can be tested to distinguish between conductive and sensorineural system loss. If the threshold is the same for air-conduction and bone-conduction tests, thresholds above 0 dB HL can be attributed to a problem in the sensorineural part of the ear. If the thresholds are different, the difference can be attributed to a problem in the conductive system (middle ear). There can also be problems in both systems. Figure 10.47a

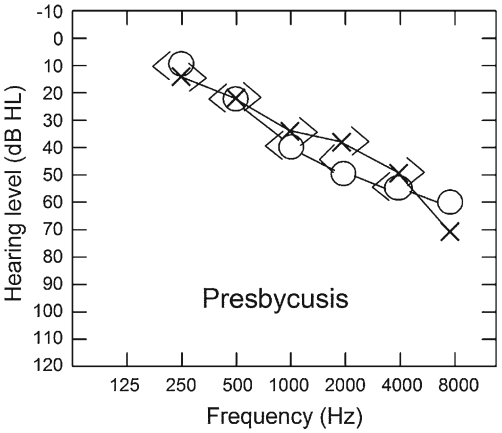


Fig. 10.45 Audiogram of a patient with bilateral hearing loss (i.e., loss in both ears) from (essentially symmetrical bilateral sloping) sensorineural hearing loss associated with presbycusis. Audiogram key: right air-conduction (*open circles*), left air-conduction (*cross symbols*), right bone-conduction (*left angular brackets*), and left bone-conduction (*right angular brackets*), all unmasked (Based on [27])

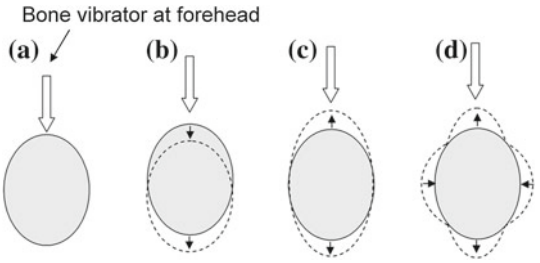


Fig. 10.46 Skull vibration patterns induced by **a** a bone vibrator positioned on the forehead, vibrating at different applied frequencies **b** 200 Hz, **c** 800 Hz, **d** 1,600 Hz. In **b–d** the vibrating skull (*dashed lines*) is shown relative to the still skull (*unbroken lines*) for a particular phase in the vibration. The small arrows depict the displacement, a displacement that is greatly exaggerated to illustrate it. In the opposing phase, the motion is reversed, with all arrows reversed (just as for a sine wave). The vibrator is off in **a** (Based on [8, 27])

shows the hearing response for someone with normal hearing in the left ear, and (unilateral, meaning in one ear) hearing loss in the right ear attributed to a problem in the right-conductive system. The sensorial system is normal, as is seen by the bone-conduction tests. Figure 10.47b shows another audiogram that indicates significant hearing loss only in one ear, the left ear, and this time it is due to hearing loss in the left sensorineural system. Here, hearing loss is seen in the left ear even with bone conduction. Note the sensorial hearing loss has a more striking dependence on frequency than the conductive hearing loss. The audiogram of a patient with hearing loss from high noise exposure (such as from too much loud rock—or classical—music) is shown in Fig. 10.48a. The loss is bilateral (in both ears) and is due to

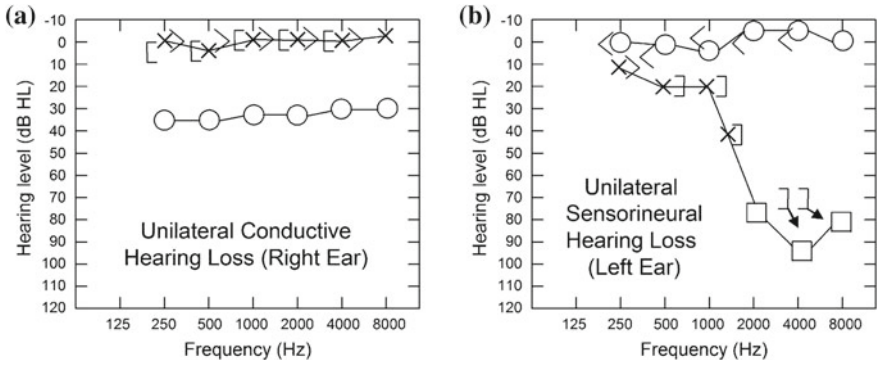


Fig. 10.47 Audiograms of patients with **a** unilateral conductive hearing loss in the right ear, with a normal left ear, and **b** unilateral sensorineural hearing loss in the left ear, with a normal right ear. Audiogram key: right air-conduction (*open circles*), left air-conduction (*cross symbols*), right bone-conduction (*left angular brackets*), and left bone-conduction (*right angular brackets*), all unmasked; left air-conduction (*open squares*), right bone-conduction (*left square brackets*), left bone-conduction (*right square brackets*), no response (*arrows*), all masked. (In masking the nontest ear is presented a constant noise signal to prevent it from detecting the signal in the test ear, so that only the test ear can respond) (Based on [27])

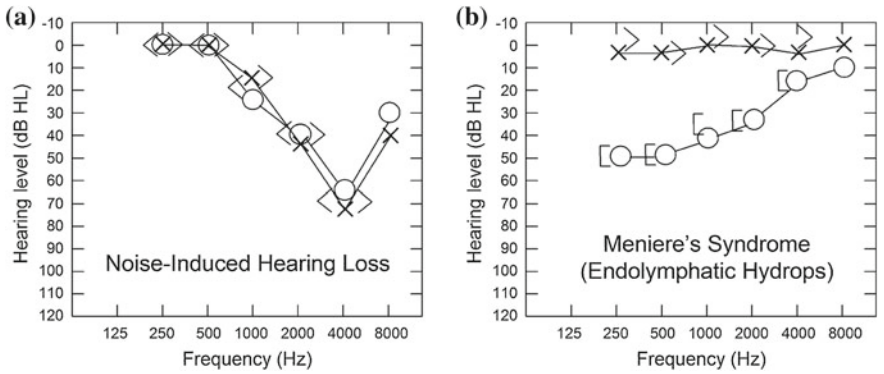


Fig. 10.48 Audiograms of patients with **a** bilateral sensorineural hearing loss associated with noise exposure and **b** Meniere's disease exhibiting low-frequency sensorineural hearing loss in the right ear. Audiogram key: right air-conduction (*open circles*), left air-conduction (*cross symbols*), right bone-conduction (*left angular brackets*), and left bone-conduction (*right angular brackets*), all unmasked; right bone conduction (*left square brackets*), masked (Based on [27])

sensorineural malfunction. In contrast to these other examples, people with Meniere's disease (an inner ear disease) show sensorineural hearing loss at low frequencies (Fig. 10.48b).

Tinnitus, which is also known as "ringing in the ears," is not uncommon. Those with severe cases can perceive a tone in one or both ears of 90 dB or louder, which is louder than the 60–70 dB of normal conversation.

10.3.2 Connections to Hearing Perception

In *psychophysics* (or *psychoacoustics*), there are four main perception attributes of basic sounds: loudness, pitch, timbre, and localization [29, 76]. Each attribute is nonlinearly related to the physical characteristics of sound, such as acoustic intensity, frequency, and distribution of frequencies:

1. *Loudness* is related to the sound intensity in a nonlinear manner. As characterized by Stevens' Law ((1.6), Table 1.15) the perception of loudness is very sublinear with stimulus. The lowest curve in Fig. 10.29 is the audibility curve, which denotes the threshold of hearing; this is the lowest intensity "equal loudness curve." This and the other "louder" "equal loudness curves" provide the dB SPL levels of acoustic intensity at each frequency (for pure tones, i.e., for single frequencies) needed to achieve the same level of perceived loudness. The degree of loudness of a sound is expressed as the dB SPL level at 1,000 Hz that has the same loudness. If this acoustic intensity at 1,000 Hz with the same loudness is called I, then the loudness in *phons* is

$$L_{\mathrm{p}}(\text{in phons}) = 10 \log \frac{I}{I_{\mathrm{ref,p}}}, \tag{10.64}$$

where $I_{\mathrm{ref,p}} = 1 \times 10^{-12}\,\mathrm{W/m^2}$. (This equation is clearly very similar to (10.10).) As seen in Fig. 10.49, above ~80 phons, approximately the same acoustic intensity is needed in low, mid, and high frequencies to perceive the same loudness in each region. Below ~80 phons, higher intensity is needed at low and high frequencies to achieve the same perceived loudness as in mid frequencies. This explains why one needs to increase the bass and treble in stereo tuners (such as by pushing the "loudness" button) at lower intensity to perceive the same, except quieter, music. Figures 10.44 and 10.45 show that older people typically have less sensitivity at high frequencies (presbycusis). Several disorders, such as Meniere's disease, also deteriorate hearing (Fig. 10.48b).

The phon scale gives no indication of relative loudness. This is provided by the unit called *sones*, which is related to phons in Fig. 10.50. A loudness sensation of 1 sone is provided by 40 dB of 1,000 Hz pure tone. A sound of 10 sones is perceived as being twice as loud as one of 5 sones. A nearly exact formulation of loudness in sones L_{s} is

$$L_{\mathrm{s}}(\text{in sones}) = \left(\frac{I}{I_{\mathrm{ref,s}}}\right)^{0.3}, \tag{10.65}$$

where $I_{\mathrm{ref,s}} = 1 \times 10^{-8}\,\mathrm{W/m^2}$ and again I is the acoustic intensity at 1,000 Hz with the same loudness [35]. (This is a little different from the near square-root scaling of perceived loudness with the acoustic intensity indicated in Table 1.15.) Another measure of loudness is the *just noticeable difference* (JND) or *difference limen* (DL), which denotes how much more intense a sound needs must be to be

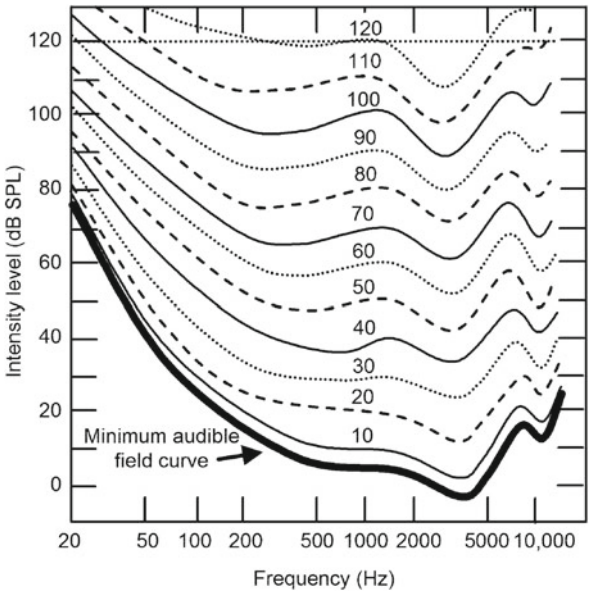


Fig. 10.49 Loudness and phon curves versus frequency with phon (loudness level) labels (Based on [27, 38, 58])

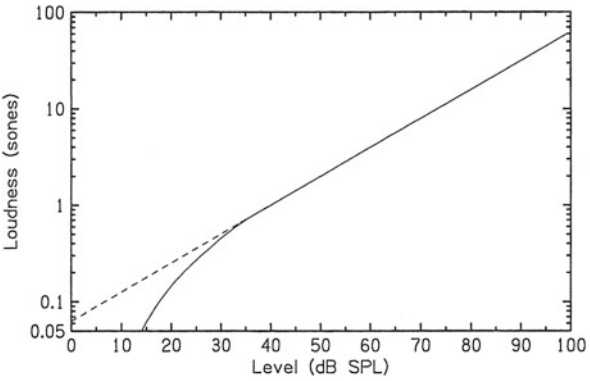


Fig. 10.50 Perceived loudness (in sones) versus loudness level of stimulus (in phons) for a 1,000-Hz tone from [67, 76], shown as the *solid line*, along with the predictions of the approximate 0.3 power law shown as the *dashed line*, which is the basis for the national and international sone scale (From [35])

perceived as being louder. For a 1,000 Hz tone, the DL for a 5 dB SPL sound is 5 dB SPL, but for 100 dB SPL it is much smaller fraction, 6 dB SPL.

2. *Pitch* denotes if a tone sounds high or low. A pressure wave that is purely sinusoidal is a pure tone; it is perceived as being low for low frequencies and high for high frequencies. The perception of pitch is expressed in units of *mels*, with the pitch of 1,000 Hz at 40 dB SPL defined to have a pitch of 1,000 mels. There is a very nonlinear relationship between mels and frequency for a pure tone (Fig. 10.51). A tone of 2,000 mels (corresponding to about 3,500–4,000 Hz) sounds like it is twice the pitch of a tone of 1,000 mels, which itself sounds twice as high as a tone of 500 mels (corresponding to 400 Hz).

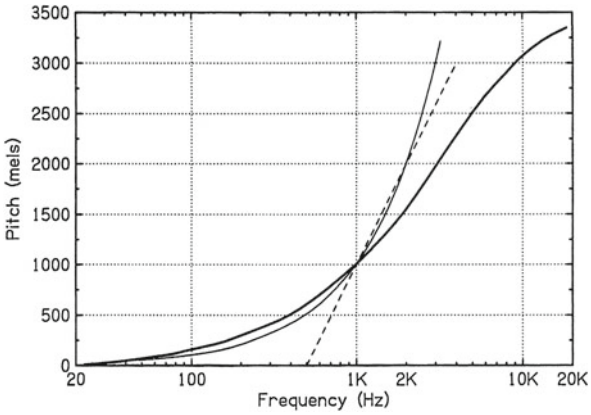


Fig. 10.51 Perceived pitch (in mels) versus frequency [68] is shown as the *heavy line* (extending from 20 Hz to almost 20 kHz), along with the *thin line* plotting pitch equal to frequency and the *dashed line* showing pitch proportional to the octave number (From [35])

There is a minimum detectable frequency change for pure tones, of about 2–3 Hz for tones below 1,000 Hz and about 1/3% fractional change for higher frequency tones. At a constant loudness of 40 phons, one can perceive 1,400 different frequencies. Also, at a constant frequency of 1,000 Hz one can perceive about 280 different levels of loudness. Consequently, people can perceive almost $1{,}400 \times 280 = 400{,}000$ different variations of pitch and loudness.

The frequency increases by a factor of two progressing from one musical note to the same note in the next octave, such as in going from A, B, C, D, E, F, G, and back to A; this continual increase in frequency with successive notes is recognized as an increase in *tone height*. In going from one octave to another, such as A_5 on a piano with a fundamental frequency at 440 Hz, to A_6 at 880 Hz, to A_7 at 1,760 Hz, and so on, there is a repetition of notes that is perceptually similar, i.e., they exhibit the same properties of *tone chroma* (here specifically here that of the note A), but with different tone heights. These notes can be graphically represented as a vertical spiral with successive notes marked; each complete octave is a loop in the spiral, and all notes with the same tone chroma (all A's, all B's, etc.) are immediately above one another in successive loops.

Most musical notes have a range of frequencies, and consist of a fundamental frequency and its harmonics. Such complex tones are perceived to be at the fundamental frequency, so a 400 Hz tone and its harmonics at 800, 1,200, 1,600 Hz, and so on are perceived to have a pitch of 400 Hz. Moreover, other complex series of frequencies are perceived to be at the minimum common difference frequency, so a complex tone with 800, 1,200, 1,600, and 2,000 Hz is perceived to have a pitch of 400 Hz, as is a complex tone with frequencies 2,000 and 2,400 Hz. These concepts are known as *periodicity pitch* or the *effect of the missing fundamental*.

Tones heard at the same time can also *mask* one other. Tones tend to mask nearby frequencies the best. Also, low-frequency tones mask high frequency tones very well, but high frequency tones do not mask low frequency tones well.

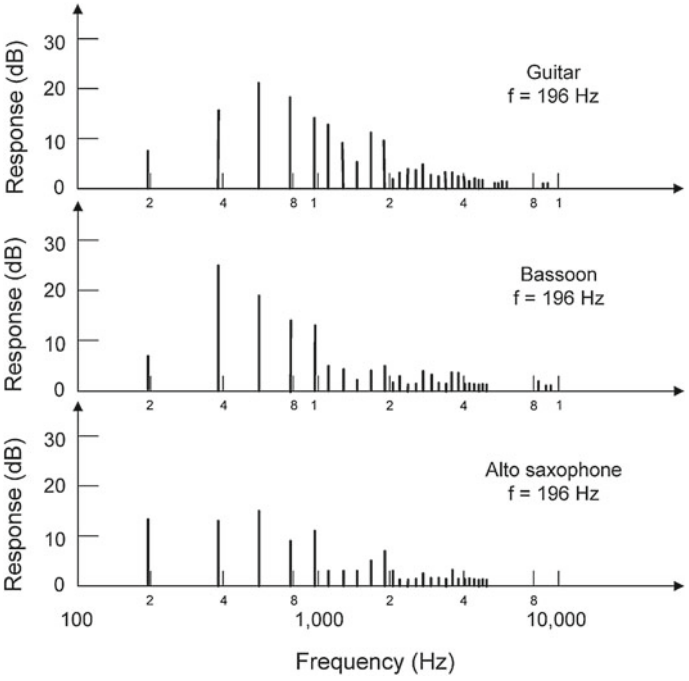


Fig. 10.52 Fourier spectra for a guitar, bassoon, and an alto saxophone playing at a fundamental frequency of 196 Hz (Based on [54])

3. Complex tones containing many frequencies can be perceived as having the same loudness and pitch, but still be perceived as being different. This is characterized as being a difference in quality or *timbre*. One reason for this is that different complex tones with the same fundamental frequency consist of harmonics with different relative intensities. These different distributions are seen in Fig. 10.52 for a 196 Hz tone from a guitar, bassoon, and alto saxophone; this explains why these different instruments sound different when the same note (fundamental frequency) is played. In fact, when musical tones of the same frequency can be played in different ways on the same instrument they sound differently, such as the 440 Hz tone played on the A and D strings of a violin, as is seen in Fig. 10.53. Moreover, perception is not only based on these distinctive harmonic structures, but also on the distinctively different buildup and decay times for the various harmonics played on a given instrument.
4. Sound is perceived as coming from a given spatial location, which is called *localization*. Binaural hearing helps people localize the origin of a sound within about 10°. The brain localizes the source on the basis of information from both ears, using the difference in arrival times—with a longer propagation time to the farther

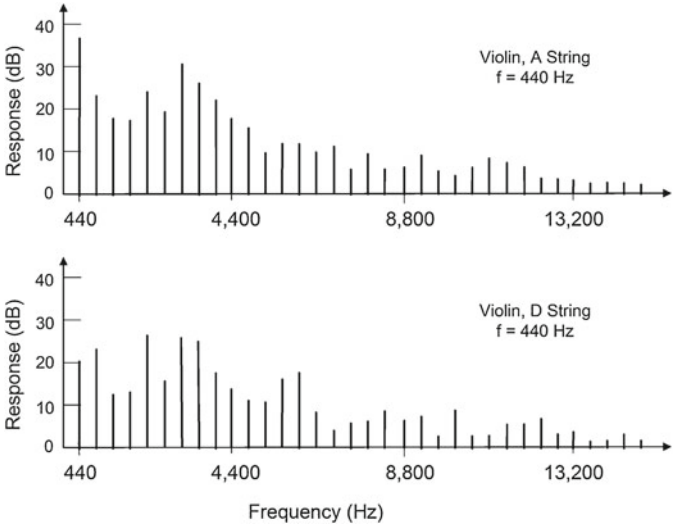


Fig. 10.53 Fourier spectra for a violin playing at a fundamental frequency of 440 Hz using the A and D strings (Based on [54])

ear—and intensity—with a weaker acoustic intensity arriving at the farther ear. This is examined further in Problem 10.79.

Perception of more complex acoustic stimuli, as in speech, is much more involved. The basic elements relate to the signal processing of the frequency components.

10.4 Other Vibrations in the Body

10.4.1 Cardiac and Other Sources of Sounds

The body can be the source of many sounds, other than from speaking, such as from chewing (noncrunchy and crunchy [56, 57, 71] foods), stomach growls, walking and running, hand clapping, hiccups, heart murmurs, body contact with anything, flapping eyelids, using towels, cracking or clicking knees and finger joints, vomiting, coughing, flatus, nose blowing, humming, whistling, sneezing, normal breathing, wheezing, and laughing. Some of these are listed in Table 10.6. The loudest of these acoustic vibrations can be sensed by the body. Some are heard and some are felt by the mechanical receptors in the body (Chap. 2). Heart sounds are described in [45] and lung sounds in [47]. Vibrations can also cause pain to be sensed in the body, as illustrated in Fig. 10.54. Knuckle cracking, and noise from other synovial joints, long thought to be due to the collapse of a cavitation bubble that is formed when the articular surfaces are being separated, may well be due to the process of forming

Table 10.6 Estimated acoustic parameters from sources in the body

Source	Maximum pressure change (atm.)	Typical frequency (Hz)
Shouting	0.05	1,000
Talking	0.005	1,000
Whispering	0.0005	1,000
Running	2.0	4
Walking	0.4	1
Clapping hands, vigorously	0.2	2
Chewing crunchy food	0.0001	1,000
Respiratory airflow turbulence	0.00004	1,000
Arterial pulse	0.02	1

From [24]

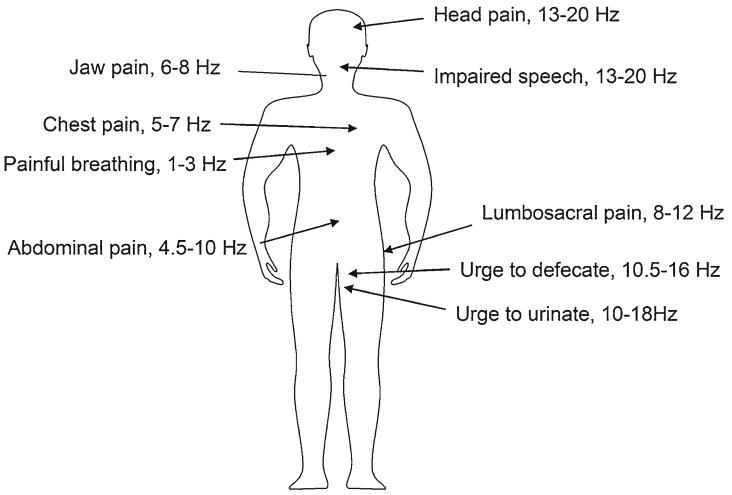


Fig. 10.54 Pain symptoms from vibrations from 1 to 20 Hz (Based on [11, 49])

a cavity [40]. By the way, though cracking releases much energy and could cause damage, frequent knuckle cracking apparently does not accelerate joint degeneration [18], despite the warnings of many generations of parents.

Auscultation is the listening to sounds made by internal organs for medical diagnosis. Modern stethoscopes have two chest pieces for such diagnostics: a "bell" chest piece applied lightly to the skin to pick up low frequency sounds and a "diaphragm" chest piece pressed firmly to the skin to minimize low frequency sound and therefore accentuate hearing high-frequency sounds.

Normally-functioning hearts make sounds that are easily heard with a stethoscope (Fig. 10.55) [45]. These are due to the opening and closing of heart valves and the flow of blood. The first heart sound (S_1) is a high-frequency sound heard in early systole due to the closing of the mitral and tricuspid atrioventricular valves, which

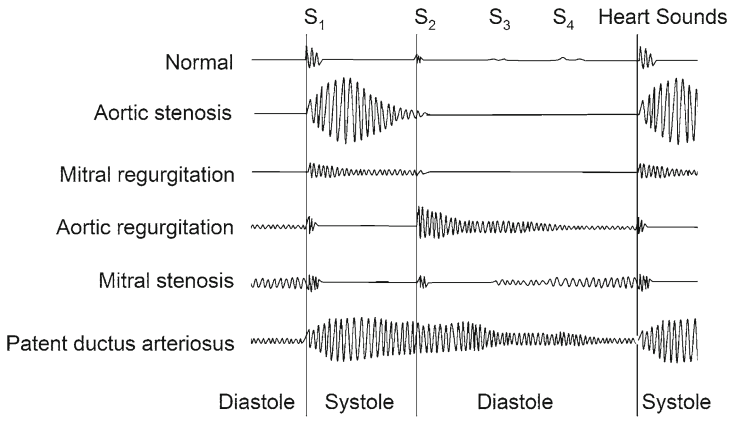


Fig. 10.55 Idealized phonocardiograms (sound traces) for normal and abnormal hearts, with left heart valve stenosis (partial blockage) or regurgitation (backflow) or with *patent ductus arteriosus*. (Patent ductus arteriosus is the condition when the ductus arteriosus (arterial shunt of pulmonary arterial blood flow from pulmonary artery to the aorta in a fetus) does not close after birth.) The times of the first (S_1), second (S_2), third (S_3), and fourth or atrial (S_4) heart sounds are shown. Also see the phonocardiogram in Fig. 8.5 (Based on [33])

occurs when the ventricular pressure exceeds the atrial pressure. The mitral valve closes about 10 ms before the tricuspid valve due to the earlier electrical excitation and contraction of the left ventricle. These different closing times are audible only when there is a disorder, such a right bundle branch block, which delays the closure of the tricuspid valve. The intensity of the S_1 sound can be accentuated or diminished by several heart disorders.

The second sound (S_2) is heard at the end of ventricular ejection and systole when the pulmonary and aortic valves close. The aortic component (A_2) precedes the pulmonary component (P_2) because the pressure gradient between the aorta and left ventricle is larger than that between the pulmonary artery and right ventricle. These two components are normally fused into one during expiration but are normally audibly split during inspiration due to the decrease in intrathoracic pressure that further delays P_2 and advances A_2. The decrease in intrathoracic pressure causes a larger pressure drop from inside to outside the pulmonary arteries and veins, and these compliance vessels increase in volume at the same internal pressure. This delay in P_2 is caused by the longer time needed to fill up the pulmonary artery. The advancing of A_2 is due to the smaller blood flow from the even-more blood-laden pulmonary veins to the left atrium and then the left ventricle, and the resulting smaller stroke volume and the smaller time needed for the left ventricle to become empty. The time of systole is approximately that between S_1–S_2 and that of diastole from S_2 to S_1 in the next cycle. These main heart sounds, S_1 and S_2, are high-frequency sounds best heard with the stethoscope diaphragm.

The third heart sound (S_3) is a dull, low-pitch sound best heard with the bell of the stethoscope, heard after opening of the atrioventricular valves during ventricular filling; it is normal in children and young adults and abnormal in middle-aged and older adults. The fourth heart sound (S_4), which usually indicates cardiac disease, is generated by an atrium contracting against a stiffened ventricle late in diastole. *Heart murmurs* are generated by the presence of turbulent blood flow rather than laminar flow, and may indicate flow across a partial obstruction, such as a stenosis (narrowing) of the aorta, increased flow rate, blood ejection into a dilated chamber, regurgitant flow (backflow) in a defective heart valve, or abnormal shunting of blood from higher-pressure to lower-pressure chambers through a membrane divider (septum) [45]. (Remember from Chap. 8 that flow in the aorta is sometimes turbulent and can be heard.) As seen in Fig. 8.5, the loudest heart sounds have frequencies mostly below 30 Hz and are therefore inaudible. The pulsatile turbulent flow heard during the measurement of blood pressure was described in Chap. 8.

10.5 Summary

The physics of sound, i.e., acoustics, describes the properties of sound waves—including how sound intensity is expressed in the units of dB—and how sound waves propagate between different media and in cavities. Acoustic models can be used to understand how sound is produced in speech, which is partly explained by the voice filtering theory, and how to characterize the human voice. They can also be used to understand sound propagation in the outer and middle ears, how nerve signals are generated in the inner ear, and modes of hearing loss. Vibrations are important elsewhere in the body and in medical diagnostics, such as in the interpretation of heart sounds and the use of ultrasound for imaging.

Problems

Sound Waves

10.1 Find the distance molecules in air move for 20 dB SPL and 120 dB SPL sound levels, both at 500 and 5,000 Hz.

10.2 A musician with perfect pitch can identify a 1 kHz pure tone in 4 ms. How many periods of the sound wave is this?

10.3 Use (10.6) and the mass density of air of 1.3×10^{-3} g/cm^3 to calculate the characteristic displacement of air for 0 dB SPL and 120 dB SPL for 4 kHz.

10.4 (a) Show that the speed of sound in air $v_{\text{air}} \propto \sqrt{T}$ and the mass density varies as $\rho_{\text{air}} \propto 1/T$, and so the acoustic impedance of air varies as $Z_{\text{air}} \propto 1/\sqrt{T}$, where T is in K.

(b) Find each of these parameters at 0, 20, and 25 °C.
(c) Does this variation of Z_{air} with T significantly affect the reflection and transmission of sound from air to the body components? Why or why not?

10.5 Find the speed of sound in water using the bulk modulus of water of 2.26 GPa. Is this the expected result?

Sound Intensity

10.6 The scale for acoustic intensity in Fig. 10.30 ranges from −20 to 140 dB SPL. What pressure range does this correspond to in dyne/cm^2?

10.7 Sound with intensity 60 dB SPL in air is incident on water. How much of it is transmitted into the water (in dB SPL and W/m^2)?

10.8 Hammering on a steel plate produces sounds that two feet away reach a maximum of 115 dB SPL. If this acoustic intensity is isotropic, what is the total power of this acoustic wave in W?

10.9 What is the change in dB if the intensity of a sound wave is:
(a) halved
(b) doubled
(c) tripled
(d) quadrupled?

10.10 Why is there is a 6 dB SPL decrease in sound level for each doubling of distance from a small isotropic source?

10.11 The acoustic intensity is 60 dB SPL at a given distance from an isotropic source. What would the intensity be if this same level source were to radiate into only 1/10 of all space?

10.12 Which sound is more intense 20 m from its isotropic source: a 10 Hz sound that is 80 dB SPL a distance 4 m from its source or a 4,000 Hz sound that is 60 dB SPL a distance 3 m from its source?

Sound Intensity for Threshold Hearing Sensitivity

10.13 (a) At what distance from an isotropic 10 μW acoustic source is the sound at the audibility threshold for a human?
(b) The hearing threshold for dogs is 1×10^{-15} W/m^2. At what distance can a dog hear this source?

10.14 (a) Express music's *tripleforte* (very loud, 1×10^{-2} W/m^2) and *triple piano* (very soft, 1×10^{-8} W/m^2) sound levels in dB SPL.
(b) At an outdoor concert with no sound amplification, the audience sitting 4 m from the orchestra hears triple forte and triple piano sounds. What are the respective acoustic intensities (in dB SPL) for those in the audience sitting 60 m away? Treat the orchestra as a point source.

10.15 (a) Show that 1 W from an isotropic acoustic power radiator produces an intensity of 115 dB SPL a distance 0.5 m from the source.
(b) Find this acoustic intensity 1 m from the source.
(c) How far can you be from the source and still barely hear it (for a 1,000 Hz source)?

10.16 Determine P_{ref} and the pressure (changes) at 100 dB SPL in mmHg.

Sound Transmission, Reflection, and Ultrasound

10.17 Table 10.3 gives the absorption coefficient α for bone as 1.6×10^{-4} s/m. Other sources give it as 14 dB/cm at 1 MHz frequency. Are these two values consistent? Why?

10.18 Can you talk through a person? (In other words, are people good acoustic shielding?) Estimate the dB loss for 3,000 Hz sound transmitted through your chest to help answer this question.

10.19 For each tissue in Table 10.3, determine the thickness of tissue needed to decrease the intensity of a 5 MHz ultrasound wave by half. (Assume losses are due only to absorption, and not due to reflection at interfaces.)

10.20 Often in ultrasound measurement the ultrasound transducer and detector are on the same probe, so reflected sound waves are detected. Determine the fraction of initial sound intensity that is detected for the following cases by tracking the beam that is transmitted through material X in the body, reflected at normal incidence from the interface of X with Y, and then again transmitted through material X. (Assume that all the sound from the transducer enters X and all leaving X after reflection from Y enters the detector. Consider attenuation in the medium, as well as reflection at the X–Y interface.):
(a) X is 1 cm of muscle and Y is bone, for 1 MHz sound
(b) X is 1 cm of muscle and Y is bone, for 10 MHz sound
(c) X is 5 cm of fat and Y is muscle, for 1 MHz sound
(d) X is 2 cm of blood and Y is muscle, for 5 MHz sound
(e) X is 1 cm of bone and Y is muscle, for 1 MHz sound.

10.21 Calculate the relative delay times between sound reflecting at the beginning of the X medium and that reflecting at the X–Y interface, for each case in Problem 10.20.

10.22 Estimate the relative delay times for 1 MHz sound reflecting normally from the first outer surface of the aorta, the first inner surface of the aorta, the second inner surface of the aorta, and second outer surface of the aorta.

10.23 When an ambulance siren blaring at a frequency f approaches you at speed v, you hear the frequency upshifted to $f' = f(1 + v/v_s)$, while when it is distancing itself from you, you hear the frequency downshifted to $f' = f(1 - v/v_s)$. This is the Doppler effect.

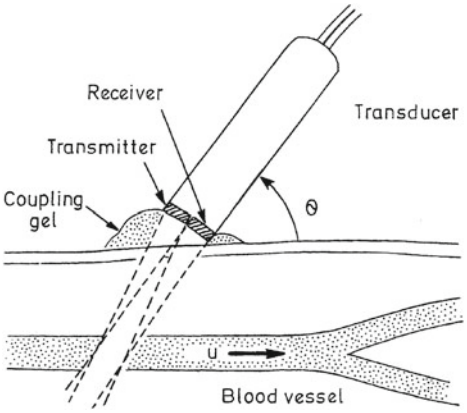


Fig. 10.56 Doppler examination of blood flow in a vessel at a speed u, with scanning shown (Reprinted from [51]. Used with permission of Elsevier)

(a) Show that this is consistent with $f' = f(1 - (v/v_s)\cos\theta)$, where θ is the angle between the velocity vector of the moving object and the position vector from you to it.
(b) Show that Doppler ultrasonography echocardiography (Fig. 10.56) can be used to determine the blood flow speed v to be

$$v = \frac{(\delta f)v_s}{2f\cos\theta}, \tag{10.66}$$

where δf is the measured Doppler shift $(f' - f)$ and v_s is the speed of sound in body tissue.
(c) Calculate the maximum Doppler shift for blood flowing in the aorta, using 1 MHz ultrasound.

10.24 A generous dab of gel is put on the ultrasound probe head before it is placed on the skin (Fig. 10.56):
(a) Why?
(b) What must be the desired acoustic properties of this gel?

10.25 Express the unit of acoustic admittance, the mmho, in SI units.

10.26 (advanced problem) With $Z = R + \mathrm{i}X_m + X_s/\mathrm{i}$, derive (10.19).

10.27 (advanced problem) With $Z = R + \mathrm{i}(X_m - X_s)$ and $Y = G + \mathrm{i}(B_m - B_s)$, derive:
(a) Equation (10.20)
(b) G, B_m, and B_s in terms of R, X_m, and X_s
(c) R, X_m, and X_s in terms of G, B_m, and B_s.

10.28 (advanced problem) With $Z = R + \mathrm{i}\omega M + S/\mathrm{i}\omega$, where ω is the radial frequency in rad/s and $\omega = 2\pi f$ where f is the frequency in Hz or cycles per second, show that

$$| Z | = \sqrt{R^2 + (2\pi f M + S/2\pi f)^2}. \tag{10.67}$$

In electronics problems, R is the resistance, M corresponds to the inductance L, and S corresponds to the reciprocal of the capacitance C ($S = 1/C$).

10.29 (advanced problem) Use (10.24) to show that the resistive term of the speed varies as a cosine wave and the inertial and stiffness terms both vary as sine waves, but with opposite signs.

10.30 The speeds of sound in the brain and skull bone are 1,550 and 4,090 m/s, respectively. What fraction of sound is lost in reflection from air to the skull bone and then from the skull bone to the brain? Assume the densities of the brain and skull bone are 1 g/cm^3.

10.31 We are usually concerned with light entering the eye, but what happens when sound enters the eye? Calculate the reflection coefficient at each interface between the air/cornea/aqueous humor/eye lens/vitreous humor. The speeds of sound in the aqueous humor, eye lens (crystalline lens), and vitreous humor are 1,510, 1,630, and 1,540 m/s, respectively. Assume that the density of each medium in the eye is 1 g/cm^3 and that the cornea and eye lens have the same properties.

10.32 The speed of sound in collagen is 3,640 m/s along the fiber axis and 2,940 m/s across this axis. What is the reflection coefficient between blood and collagen, for sound traveling in both directions in the collagen?

Speaking

10.33 The oral cavity of a child is 8 cm long, as measured from her lips to vocal folds. What is the fundamental oscillation frequency of this cavity? (Treat the oral cavity as a cylinder open at one end and closed on the other.) Does this make sense in light of the differences of the voices of children and adults?

10.34 Plot on the same set of axes all of the resonant frequencies below 5 kHz for an 18-cm long cylinder that is open on both ends and a 16-cm long cylinder that is open on one end and closed on the other.

10.35 Smokers may have vocal folds that are slightly swollen and inflamed, and therefore, perhaps have folds that are more massive than those of nonsmokers. How would this affect the vibration frequency of their vocal folds? Would the voices of smokers be relatively deeper or higher pitched?

10.36 Explain why people speak in a high pitch after taking a breath of helium. The speed of sound in helium is about 970 m/s. (Show this, given $\gamma = c_p/c_v$ is 5/3 for helium.)

10.37 In the text, the inverse relationship between the vocal-fold frequency and vocal-fold length was explained by considering the resonant frequencies of an oscillating string. Use (4.4) to show that this relationship is also expected if the vocal folds are modeled as a free, spring like object oscillating length-wise.

10.38 Estimate the vibration frequency of the vocal folds, by assuming they are a spring-like object oscillating length-wise that is 1 cm long, 0.3 cm wide, and 0.3 cm thick, with a Young's modulus of 100 kPa, and a mass density of 1 g/cm^3 [1]. (See (4.4).)

10.39 We can describe the production of the "m" sound as being voiced bilabial nasal. In what way does the production of the "n" sound differ?

10.40 Compared to men, do women use higher or lower harmonics of their fundamental buzzing frequency to produce the same vowel formant? Why?

10.41 Use the vowel formant plot (Fig. 10.19) to sketch the transmission curve of the vocal tract for three vowels.

10.42 Describe the frequency spectra of the vowels and consonants in Fig. 10.9. Point out their similarities and differences.

10.43 Plot the first- and second-formant frequencies for each vowel in Fig. 10.9 on the same set of axes.

10.44 Plot the first- and second-formant frequencies for the vowel in Fig. 10.20 on the same set of axes.

10.45 (advanced problem) Explain why the two-tube model in Fig. 10.25b, d, f, explains the mode shifts for the "ee" sound as in "see," which are shifted from the predictions of the one-tube model in Fig. 10.24a.

Hearing Mechanism

10.46 What is the resonant frequency of the 1.3 cm long ear canal of a baby? How does it compare to that of an adult?

10.47 The amplitude of motion of the eardrum is 0.03 μm when measured at 100 dB SPL for 3,000 Hz. Assuming the amplitude is linear with the total force on the eardrum (which is a good assumption for pressures below 130 dB SPL), find the amplitude of eardrum motion at 0 dB SPL in m.

10.48 Model the motion of the eardrum as a flat membrane that is fixed at its ends, with a displacement that increases linearly from the edge to the center of the eardrum. (The motion is really more closely sinusoidal than this triangular mode shape.) If this maximum displacement at the center is 0.03 μm, as in Problem 10.47, find the full change of angle the eardrum makes during its motion (in radians and degrees).

10.49 Compare the potential energy of motion of the eardrum at 3,000 Hz to that at 1 Hz for the same amplitude of motion. (Assume the motion can be modeled as a simple harmonic oscillator.)

10.50 Is modeling the eardrum as a freely vibrating object reasonable, given that the stapes touches it? Why?

10.51 Do people have good auditory sensitivity at the fundamental frequencies of their voices? Is this important? Why?

10.52 What is the gain in dB due to the middle ear if only the force enhancement in the middle ear (and not the entire torque enhancement) is considered?

10.53 Sketch the hair cell responses in Fig. 10.39 on a linear–linear plot of acoustic intensity versus frequency.

10.54 How are the three sharp hair cell responses in Fig. 10.39 related to the responses for a oscillator as shown in Figs. 10.7 and D.3?

10.55 Design hair cells composed of keratin with resonant frequencies ranging from 20 Hz to 20 kHz with:
(a) A fixed length of 2 mm
(b) A fixed diameter of 20 μm or
(c) Dimensions of 2 mm length and 20 μm diameter for the hair resonant at 1 kHz, with variations in these two parameters that cause equal changes in frequency for different hair cells.

10.56 Calculate the fundamental frequency for lateral vibrations of a rod of solid bone of length 10 mm and diameter 1 mm ($Y = 1 \times 10^{11}$ Pa and density $\rho = 3{,}000\,\text{kg/m}^3$) fixed at one end to an incompliant base.

10.57 (a) Show that 1 $\text{mmH}_2\text{O} = 0.98$ daPa.
(b) Express the tympanogram range of +200 daPa to −300 daPa in mmH_2O.

10.58 Show that the impedance relationship equivalent to the admittance relationship for tympanograms, $Y_{\text{middle ear}} = Y_{\text{total}} - Y_{\text{outer ear}}$, is $Z_{\text{middle ear}} = Z_{\text{outer ear}} Z_{\text{total}} / (Z_{\text{outer ear}} - Z_{\text{total}})$.

10.59 (advanced problem) For adiabatic conditions (no heat flow) PV^{γ} is a constant. Show that (10.59) becomes $\Delta P = -\gamma P(\Delta V)/V$ and that this leads to (10.61).

Hearing Levels and Perception

10.60 A student in a class wants to set the tone of his cell phone ringer so that he could hear it, but his instructor, who is a bit older than he is, cannot. Would a suitable fundamental frequency be 250, 1,000, 17,000, or 30,000 Hz? Why?

10.61 The acoustic power incident on the eardrum at threshold (0 dB SPL) is equivalent to how many optical photons [52]? (Hint: Use the known threshold intensity and the dimensions of the eardrum.)

10.62 Compare the intensity of the crack of a bat hitting a baseball as heard by the catcher and by a fan in the bleachers.

10.63 A noisy elevated train (in the open air) in Brooklyn in New York City causes acoustic discomfort to those 5 ft away from it. How many city blocks away can it be heard? (Treat the train as a point source, even though this is clearly an approximation. There are 20 canonical city blocks in a mile. Also, remember that background noises in the city could mask the sound of the train, so the threshold intensity for hearing is much above the threshold of hearing pure tones in a quiet room.)

10.64 The explosion of 23 kg of TNT creates a sound level of 200 dB SPL a distance 3 m away from the detonation. Assume the total acoustic energy produced in such explosions is proportional to the mass of the TNT:
(a) How far away must you be to avoid the threshold of pain from a blast from 100 kg of TNT?
(b) What fraction of the energy released from the detonation is in the form of acoustic energy over the hemispherical release region? Assume the blast is 50 ms long and a ton of TNT releases 4.18×10^9 J.
(c) It has been reported that 1 ton of TNT produces 120 dB SPL at 15 km. Is this consistent with the data given in this problem?

10.65 A static pressure of 8×10^3 Pa across the eardrum can cause it to rupture [11]. How does this compare to the sound pressure from a 160-dB SPL sound that can also cause the eardrum to rupture?

10.66 According to Fig. 10.30, what range of frequencies is needed to hear speech and over what overall volume range in dB SPL is this required?

10.67 According to Fig. 10.30, what range of frequencies is needed to hear orchestral music and over what overall volume range in dB SPL is this required?

10.68 Resketch the normal hearing range in Fig. 10.30 along with the music range for rock and roll music (which did not exist in 1934 when this curve was made).

10.69 Show that (10.65) is referenced to a loudness corresponding to 40 phons.

10.70 Show that the loudness in phons, L_p, and sones, L_s, are related by: $L_p = 33.3 \log L_s + 40$, by using (10.64) and (10.65).

10.71 Which is loudest and which is the most quiet for these three sounds: one at 7,000 Hz with loudness of 60 phons, one at 4,000 Hz with a loudness of 8 sones, or one at 1,000 Hz with intensity 50 dB SPL? (As part of this problem, express each of the three sets of data into phon and sone loudness units.)

10.72 Which 1,000 Hz sound is louder 50 ft from its isotropic source: one with an intensity 80 dB SPL a distance 5 ft from the source or one that is 65 dB SPL a distance 25 ft from the source? (In both cases, calculate the intensity in dB SPL at 50 ft.)

10.73 Approximately 80 dB SPL is needed to achieve a loudness of 80 phons at 100 and 1,000 Hz. To achieve 40 phons, 40 dB SPL is needed at 1,000 Hz. How much more acoustic intensity is needed to attain this loudness at 100 Hz? Express your answer in dB and by the factor increase in intensity needed at 100 Hz relative to 1,000 Hz.

10.74 You have a radio with inexpensive speakers that produce sound only in the 250–5,000 Hz frequency range. Will you be able to hear notes with 100 Hz fundamental frequencies, and if so, why?

10.75 Does the relative loudness scale given by sones agree with the power law dependence described in Table 1.14?

10.76 You want to design an organ with a 55 Hz tone:
(a) Show that you need to use a 3-m long length of pipe (open at both ends) to produce this resonant frequency.
(b) Say this 3-m long pipe is too long to use in the organ, but you still want to perceive a 55-Hz note tone. Explain why you can play notes from pipes that are 1.0 and 1.5 m long at the same time and hear a 55-Hz tone.

10.77 The musical scale of "Just Intonation" consists of tones that sound pleasing when sounded together or immediately after one another. Such pleasing combinations occur when the notes are harmonics of each other or have frequencies that are related by fractions with relative small integral numerators and denominators [54]. The frequencies of the notes in one octave in the Tonic C in the major scale of Just Intonation are f for a C tone, $9f/8$ for D, $5f/4$ for E, $4f/3$ for F, $3f/2$ for G, $5f/3$ for A, $15f/8$ for B, and $2f$ for C:
(a) Find the frequencies in this scale if the A tone has a frequency of 440 Hz.
(b) Calculate the lengths of pipes with fundamental frequencies at the frequencies of each of these notes (with the tubes open on both ends).

10.78 In the text, the difference limen for a 1,000 Hz tone was cited to be 5 dB SPL for a 5 dB SPL sound and a much smaller fraction, 6 dB SPL, for the much louder 100 dB SPL sound. Determine and then compare the absolute increase in acoustic intensity (in W/m^2) in both cases.

10.79 A person, 2 m from your left ear, speaks to you, and his voice reaches your left ear with an intensity 60 dB SPL:
(a) If your right ear receives his voice delayed by 0.05 ms, where is this speaker located relative to you? (Say a person directly in front of you is at 0° and immediately to your right is at 90°. Assume your ears are separated by 20 cm.)
(b) If the intensity of the voice decreases inversely as the square of the propagation distance, how much lower (in dB) is the sound arriving in the right ear?

References

1. B.K. Ahlborn, *Zoological Physics: Quantitative Models, Body Design, Actions and Physical Limitations in Animals* (Springer, Berlin, 2004)
2. R.M.N. Alexander, *The Human Machine* (Columbia University Press, New York, 1992)
3. J.B. Allen, S.T. Neely, Micromechanical models of the cochlea. Phys. Today 40–47 (1992)
4. L.H. Arnal, A. Flinker, A. Kleinschmidt, A.-L. Giraud, D. Poeppel, Human screams occupy a privileged niche in the communication soundscape. Curr. Biol. **25**, 1–6 (2015)
5. C. Avendano, L. Deng, H. Hermansky, B. Gold, in Speech Processing in the Auditory System, eds. by S. Greenberg, W.A. Ainsworth, A.N. Popper, R.R. Fay, The Analysis and Representation of Speech (Springer, Berlin, 2004), Chap. 2, pp. 63–100
6. P. Avery, S. Ehrlich, *Teaching American English Pronunciation* (Oxford University Press, New York, 1992)
7. M.J. Ball, J. Rahilly, *Phonetics: The Science of Speech* (Arnold, London, 1999)
8. G. Bekesy, Zur Theorie des Horens bei der Schallauftiahme durch Knochenleitung. Ann. Phys. **13**, 111–136 (1932)
9. G. von Békésy, *Experiments in Hearing* (McGraw-Hill, New York, 1960)
10. Bell Labs, Record **12**(6), 314 (1934)
11. J.R. Cameron, J.G. Skofronick, R. Grant, *Physics of the Body*, 2nd edn. (Medical Physics, Madison, 1999)
12. G. Clark, *Cochlear Implants: Fundamentals and Applications* (Springer, Berlin, 2003)
13. E.A. Culler, J.D. Coakley, K. Lowy, N. Gross, A revised frequency-map of the guinea-pig cochlea. Am. J. Psychol. **56**, 475–500 (1943)
14. J.T. Cushing, The spring-mass system revisited. Am. J. Phys. **52**, 925–933 (1984)
15. P. Dallos, A.N. Popper, R.R. Fay (eds.), *The Cochlea* (Springer, Berlin, 1996)
16. R. Daniloff, G. Schuckers, L. Feth, *The Physiology of Speech and Hearing: An Introduction* (Prentice-Hall, Englewood Cliffs, 1980)
17. P.B. Denes, E.N. Pinson, *The Speech Chain: The Physics and Biology of Spoken Language*, 2nd edn. (W.H. Freeman and Co., New York, 1993)
18. K. Deweber, M. Olszewski, R. Ortolano, Knuckle cracking and hand osteoarthritis. J. Am. Board Fam. Med. **24**, 169–174 (2011)
19. R.L. Diehl, B. Lindblom, Explaining the structure of feature and phoneme inventories: the role of auditory distinctiveness. in Speech Processing in the Auditory System, eds. S. Greenberg, W.A. Ainsworth, A.N. Popper, R.R. Fay (Springer, Berlin, 2004), Chap. 3, pp. 101–162
20. G.L. Fain, *Sensory Transduction* (Sinauer, Sunderland, 2003)
21. G. Fant, *Acoustic Theory of Speech Production* (Mouton, The Hague, 1970)
22. R. Fettiplace, C.M. Hackney, The sensory and motor roles of auditory hair cells. Nat. Rev. Neurosci. **7**, 19–29 (2006)
23. N.H. Fletcher, *Acoustic Systems in Biology* (Oxford University Press, New York, 1992)
24. R.A. Freitas Jr., Nanomedicine, Volume I: Basic Capabilities (Landes Bioscience, Austin, 1999)
25. B.O. French, Appraisal of Apollo launch noise. Aerosp. Med. **38**, 719–722 (1967)
26. S.A. Gelfand, *Hearing: An Introduction to Psychological and Physiological Acoustics* (Dekker, New York, 1981)
27. S.A. Gelfand, *Essentials of Audiology*, 2nd edn. (Thieme, New York, 2001)
28. D.C. Giancoli, *Physics*, 4th edn. (Prentice-Hall, Englewood Cliffs, 1995)
29. E.B. Goldstein, *Sensation and Perception*, 4th edn. (Brooks/Cole, Pacific Grove, 1996)
30. D.M. Green, *An Introduction to Hearing* (Lawrence Erlbaum, Hillsdale, 1976)
31. S. Greenberg, W.A. Ainsworth, A.N. Popper, R.R. Fay (eds.), *Speech Processing in the Auditory System* (Springer, Berlin, 2004)
32. J. Gruber, Basic multifrequency tympanometry: the physical background. www.lymenet.de/symptoms/tympanom/basictym.htm
33. A.C. Guyton, J.E. Hall, *Human Physiology and Mechanisms of Disease*, 6th edn. (Saunders, Philadelphia, 1997)

34. C.M. Hackney, D.H. Furness, Mechanotransduction in vertebrate hair cells: structure and function of the stereociliary bundle. Am. J. Physiol. Cell. Physiol. **268**, 1–13 (1995)
35. W.M. Hartmann, *Signals, Sound, and Sensation* (Springer, Berlin, 1998)
36. H.L. Hawkins, T.A. McMullen, A.N. Popper, R.R. Fay (eds.), *Auditory Computation* (Springer, Berlin, 1996)
37. D.C. Hodge, G.R. Garinther, in Noise and Blast, eds. J.F. Parker Jr., V.R. West, Bioastronautics Data Book (NASA, Washington, 1973), Chap. 15, pp. 693–750
38. International Standards Organization (ISO): Recommendation 226: Normal Equal Loundness Contours for Pure Tones and Normal Thresholds of Hearing Under Free-Field Listening Conditions (ISO, Geneva, 1961)
39. J.F. Kennedy, Space Center. Results of AS-503 facilities and environmental measurements, Vol. 1. Technical. Report 927 (1969)
40. G.N. Kawchuk, J. Fryer, J.L. Jaremko, H. Zeng, L. Rowe, R. Thompson, Real-time visualization of joint cavitationn. PLOS ONE. (2015). doi:10.1371/journal.pone.01194701
41. J. Keener, J. Sneyd (eds.), *Mathematical Physiology II: Systems Physiology*, 2nd edn. (Springer, Berlin, 2009)
42. C.P. Lebo, R.C. Reddell, The presbycusis component in occupational hearing loss. Laryngoscope **82**, 1399–1409 (1972)
43. P. Lieberman, *Toward an Evolutionary Biology of Language* (MA, Belknap, Cambridge, 2006)
44. D.E. Lieberman, *The Story of the Human Body: Evolution, Health, and Disease* (Pantheon, New York, 2013)
45. L.S. Lilly, Heart sounds and murmurs, in Pathophysiology of Heart Disease: A Collaborative Project of Medical Students and Faculty, 3rd edn., ed. by L.S. Lilly (Lippincott Williams & Wilkins, Philadelphia, 2003), Chap. 2, pp. 29–43
46. S.E. Linville, The sound of senescence. J. Voice. **10**, 190–200 (1996)
47. R.G. Loudon, R.L.H. Murphy, Lung sounds, in The Lung: Scientific Foundations, 2nd edn., ed. by R.G. Crystal, J.B. West, P.J. Barnes, E.R. Weibel (Lippincott-Raven, New York, 1997), Chap. 100, pp. 1383–1391
48. T.J. Lynch III, V. Nedzelnitsky, W.T. Peake, Input impedance of the cochlea in cat. J. Acoust. Soc. Am. **72**, 108–130 (1982)
49. E.B. Magid, R.R. Coermann, G.H. Ziegenruecker, Human tolerance to whole body sinusoidal vibration. Aerosp. Med. **31**, 921 (1960)
50. R.H. Margolis, Acoustic immittance measurements, in Audiology: Diagnosis, ed. by R.J. Roeser, M. Valente, H. Hosford-Dunn (Thieme, New York, 2000), Chap. 17, pp. 381–423
51. W.N. McDicken, *Diagnostic Ultrasonics: Principles and Use of Instruments*, 3rd edn. (Churchill Livingstone, Edinburgh, 1991)
52. H.J. Metcalf, *Topics in Classical Biophysics* (Prentice-Hall, Englewood Cliffs, 1980)
53. H.F. Olson, *Acoustical Engineering* (Van Nostrand, Princeton, 1957)
54. H.F. Olson, *Music, Physics, and Engineering*, 2nd edn. (Dover, New York, 1967)
55. G.E. Peterson, H.L. Barney, Control methods used in a study of vowels. J. Acoust. Soc. Am. **24**, 175–184 (1952)
56. M. Roach, *Gulp: Adventures on the Alimentary Canal* (Norton, New York, 2013)
57. Mary Roach on studying how humans chew and eat. N.Y. Times (2013). http://www.nytimes.com/2013/03/26/health/mary-roach-on-studying-food-and-how-humans-eat-it.html
58. D.W. Robinson, R.S. Dadson, A re-determination of the equal loudness relations for pure waves. Br. J. Appl. Phys. **7**, 166–181 (1956)
59. D.W. Robinson, G.J. Sutton, Age effect in hearing: a comparative analysis of threshold data. Audiology **18**, 320–334 (1979)
60. R.J. Roeser, M. Valente, H. Hosford-Dunn (eds.), *Audiology: Diagnosis* (Thieme, New York, 2000)
61. R.T. Sataloff, D. Caputo, Rosen, M. Hawkshaw, J.R. Spiegel, The aging adult voice. J. Voice. **11**, 156–160 (1997)
62. R.F. Schmidt, F. Lang, G. Thews, *Physiologie des Menschen*, 29th edn. (Springer, Berlin, 2005)

63. E.D. Schubert, *Hearing: Its Function and Dysfunction* (Springer, Berlin, 1980)
64. H. Spoendlin, The afferent innervation of the cochlea, in *Electrical Activity of the Auditory Nervous System*, ed. by R.F. Naunton, C. Fernandez (Academic, London, 1978), pp. 21–41
65. A. Spoor, Presbycusis values in relation to noise-induced hearing loss. Int. Audiol. **6**, 48–57 (1967)
66. K.N. Stevens, *Acoustic Phonetics* (MIT, Cambridge, 1998)
67. S.S. Stevens, H. Davis, Hearing: Its Psychology and Physiology (Wiley, New York, 1938), Reprinted by the Acoustical Society of America (1983)
68. S.S. Stevens, J. Volkman, The relation of pitch to frequency. Am. J. Psych. **53**, 329–353 (1940)
69. I.R. Titze, *Principles of Voice Production* (Prentice-Hall, Englewood Cliffs, 1994)
70. J. Tonndorf, S.M. Khanna, The role of the tympanic membrane in middle ear transmission. Ann. Otol. Rhinol. Laryngol. **79**, 743–753 (1970)
71. T. van Vliet, *Rheology and Fracture Mechanics of Foods* (CRC Press, Boca Raton, 2013)
72. S. Webb (ed.), *The Physics of Medical Imaging* (Adam Hilger, Bristol, 1988)
73. D.A. Winter, *Biomechanics and Motor Control of Human Movement*, 3rd edn. (Wiley, New York, 2005)
74. A.B. Wolbarst, *The Physics of Radiology* (Appleton & Lange, Norwalk, 1993)
75. W.A. Yost, *Fundamentals of Hearing: An Introduction*, 3rd edn. (Academic, San Diego, 1994)
76. E. Zwicker, H. Fastl, *Psychoacoustics: Facts and Models* (Springer, Berlin, 1990)